Journal of Water Process Engineering 59 (2024) 105065

Contents lists available at ScienceDirect

Journal of Water Process Engineering

journal homepage: www.elsevier.com/locate/jwpe

A critical review of breakthrough models with analytical solutions in a fixed-bed column

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ARTICLE INFO

Editor: Xiaohong Guan

Keywords:
Adsorption
Fixed bed
Breakthrough curve
Fractal-like kinetics
Controversy

ABSTRACT

The modeling and simulation of the breakthrough curves in a fixed-bed column play a vital role in evaluation of the adsorption capability and prediction of the dynamic adsorption behaviors. With the popular use of the breakthrough models, the relevant mistakes and inconsistencies also rise and are repeated in subsequent publications. This work conducts a profound review of the fundamental principles of continuous adsorption and the underlying assumptions, curve characteristics, intrinsic relationships and application scope of the widely used breakthrough models with analytical solutions. Besides, the physical meanings of the model parameters are clarified, the relevant mistakes and controversies in the fixed-bed studies are addressed and the reasons for the asymmetric breakthrough curves are discussed. Error statistics and residual plot are a good measure of the goodness of fit. The *F*-test and Akaike's information criterion (AIC) can be used to compare the fitting results from two models for the same dataset. The fractal-like kinetics provides new insights into the diffusion-limited adsorption process in a heterogeneous system. Some empirical breakthrough models are also considered to be an alternative method to description of the dynamic adsorption behaviors. This review aims to help beginners better understand and use the breakthrough models.

1. Introduction

Adsorption is a phase transfer process that means the enrichment of chemical species on the surface of the porous adsorbents [1]. It has become an indispensable unit operation to eliminate impurities or undesirable substances and recover compounds with high added value from municipal and industrial wastewaters [2]. The engineered adsorption processes more adopt the fixed bed rather than a batch reactor since the former has greater advantages such as simple mode of operation, efficient utilization of the adsorbent capacity, suitable treatment of large volumes, and ease of scaling up [3–5]. Compared with the traditional design-build-test approach, the design and optimization of an adsorption system increasingly rely on the mathematical modeling and simulation to save cost and time [6]. It is desirable to carry out the mathematical modeling of mass transfer, chemical reaction and thermodynamic equilibrium during adsorption process [7]. The rigorous prediction of the breakthrough curve contributes to understanding the dynamic adsorption behaviors and transport properties and avoiding the extensive experiments that tend to be expensive and time-consuming [8]. The accuracy of prediction is related to the quality and availability of the experimental data and to the underlying assumptions and rational approximations of the mathematical models.

It is inherently difficult to develop a mathematical model to accurately describe the dynamic adsorption behaviors in a fixed-bed column because the concentration profiles are a function of both space and time in the liquid and solid phases [9]. The phenomenological model established by the mass balance has the potential to identify mechanisms related to the mass transfer and the ability to make predictions outside of the scope of evaluation for obtaining the model parameters [10]. Although more general and mathematically rigid, it requires complicated numerical solutions [11]. Under initial and boundary conditions, the complete analytical solution of partial differential equations involved in phenomenological model is not available. If the goal is to accurately predict the breakthrough behaviors in a fixed-bed column, the use of simpler and more tractable models that avoid the need for

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https://doi.org/10.1016/j.jwpe.2024.105065
Received 16 November 2023; Received in revised form 10 February 2024; Accepted 23 February 2024
Available online 29 February 2024

Nomenclature

a_0	weight of solute uptake per unit volume of the bed (mg L^{-1})
c_0	concentration of the solute at the inlet of column (mg L^{-1})
c_b	effluent concentration at the breakthrough time (mg L^{-1})
c_e	concentration of the solute at equilibrium (mg L^{-1})
D_L	axial dispersion coefficient (cm^2 min^{-1})
d_p	adsorbent particle diameter (cm)
k_{BA}	Bohart–Adams rate constant (L mg^{-1} min^{-1})
K_F	Freundlich constant (mg g^{-1} $min^{-1/n}$)
K_f	overall mass-transfer coefficient in the liquid phase (cm min^{-1})
$K_f a$	volumetric mass-transfer coefficient (min^{-1})
K_L	Langmuir constant (L^{-1} mg)
k_s	mass-transfer coefficient for solid phase (min^{-1})
k_T	Thomas rate constant (mL mg^{-1} min^{-1})
K_T	mass-transfer coefficient for liquid phase (min^{-1})
k_{YN}	Yoon–Nelson rate constant (min^{-1})
q_0	equilibrium loading of the bed or adsorption capacity at saturation (mg g^{-1})
q_e	amount of the solute uptake at equilibrium (mg g^{-1})
q_{max}	maximum adsorption capacity (mg g^{-1})
q_t	amount of the solute uptake at time t (mg g^{-1})
$t_{1/2}$	operating time at $c/c_0 = 0.5$ (min)
t_b	breakthrough time (min)
t_s	saturation time (min)
V_{bed}	volume of bed (L)
β_0	external mass-transfer coefficient (min^{-1})
β_a	effective kinetic coefficient (min^{-1})
a	mass-transfer area per unit volume of the bed (cm^{-1})
A_0	fractal-like Clark rate constant (dimensionless)
c	concentration of the solute at the outlet of column (mg L^{-1})
D_m	molecular diffusivity calculated by the Stokes–Einstein equation (cm^2 min^{-1})
h	fractal-like exponent (dimensionless)
k_0	fractal-like rate constant
k_n	n-order Bohart–Adams rate constant (L^n mg^{-n} min^{-1})
$k_{BA,0}$	fractal-like Bohart–Adams rate constant (L mg^{-1} min^{h-1})
$k_{T,0}$	fractal-like Thomas rate constant (mL mg^{-1} min^{h-1})
$k_{YN,0}$	fractal-like Yoon–Nelson rate constant (min^{h-1})
m	weight of the adsorbent in the bed (g)
n	Freundlich constant (dimensionless)
p	number of the model parameters
t	operating time (min)
u	flow rate per unit cross-sectional area of column (cm min^{-1})
v	flow rate (mL min^{-1})
V	volume of the effluent at the outlet (mL)
v_d	linear velocity of movement of the adsorption zone (cm min^{-1})
x	bed height (cm) or relative concentration ($x = c/c_0$)
z	axial position of bed (cm)
ε	void fraction of bed (dimensionless)
μ	migration velocity of the concentration front in the bed (cm min^{-1})
ρ	bulk density of the bed (g L^{-1})
σ_a	standard deviation of the position of the adsorption zone front (cm)
τ	operating time required to reach 50 % breakthrough (min)

numerical solutions appears more suitable and logical [12]. Analytical solutions can clearly present the dependence of the process on the operating parameters in a way not possible with numerical solutions [13]. To this end, it is desirable to develop simple mathematical models to satisfactorily predict the dynamic adsorption behaviors. The Bohart–Adams [14], Thomas [15], Yoon–Nelson [16] and Clark [17] models are frequently applied for modeling of the breakthrough curves since they have the simple analytical solutions and can be easily linearized, allowing their free parameters to be estimated by the linear fitting [18]. Despite different underlying assumptions or empirical approximations, these mathematical models have achieved great success in the modeling of the experimental data from adsorption of heavy metals [19] and emerging contaminants such as antibiotics [20] and microplastics [21].

However, with the popular propagation of these models, the relevant problems also come along. The breakthrough models are blindly selected to analyze the experimental data probably due to a lack of the understanding of the fundamental principles of continuous adsorption as well as the underlying assumptions, curve characteristics, intrinsic relationships and application scope of the breakthrough models, resulting in a poor fit or even incorrect conclusions [22,23]. The main limitation of these models is the assumption of the time-independent rate constant or transport properties, not interpreting the heterogeneous porous structures of the adsorbent and the changes in diffusive microenvironments caused by the progressive occupation of the adsorption sites by the adsorbate molecules [24]. Moreover, a major difficult for the use of the breakthrough models is how to accurately estimate the model parameters. A common way to evaluate the goodness of fit is to adopt various error statistics such as the coefficient of determination (R^2) [25], chi-squared value (χ^2) [26] and root of mean squared error (RMSE) [27]. But, these error statistics are not adequate criteria but the residual plot [28]. The selection of an appropriate model based on both error statistics and residual plot is relatively rare in fixed-bed studies [29]. The fitting results from different models for the same dataset have not been quantitatively compared yet. This work attempts to address these problems and controversies to help beginners to better understand the breakthrough models.

2. General description of fixed bed

Continuous adsorption is a non-stationary rate-controlled process in a fixed-bed column, i.e. a time- and distance-dependent process [30]. The fixed bed is one of the basic forms of the dynamic operations in the adsorption field. Compared with the batch operation, the main advantage of the fixed-bed adsorption is to more conveniently scale up the adsorption process by the dimensional similarity approach (i.e. geometric and kinematic similitude). The fixed-bed adsorber through scaling up can be applied to the treatment of municipal and industrial wastewaters. The fixed-bed adsorption can provide reliable information on the breakthrough time, loss of adsorption capacity during subsequent cycles and acceptable flow rate [31]. However, it also faces some challenges such as high pressure drop, channeling effects, non-ideal flow behaviors and mass-transfer resistances, limiting its application in wastewater treatment [32].

As shown in Fig. 1, the adsorption process takes place when the influent moves through the bed. The bed is divided into three regions: saturation zone, mass-transfer zone (MTZ) and adsorption zone. In the saturation zone, the adsorption of the contaminant reaches the dynamic equilibrium at the solid/solution interface, where the amount adsorbed q_0 is in equilibrium with the influent concentration c_0. The available capacity of the adsorbent is exhausted and no mass transfer occurs from solution to the adsorbent particles. In the adsorption zone, the adsorbent particles are not loaded by the adsorbate molecules, and the concentration of the contaminant is equal to zero. Adsorption occurs only in the

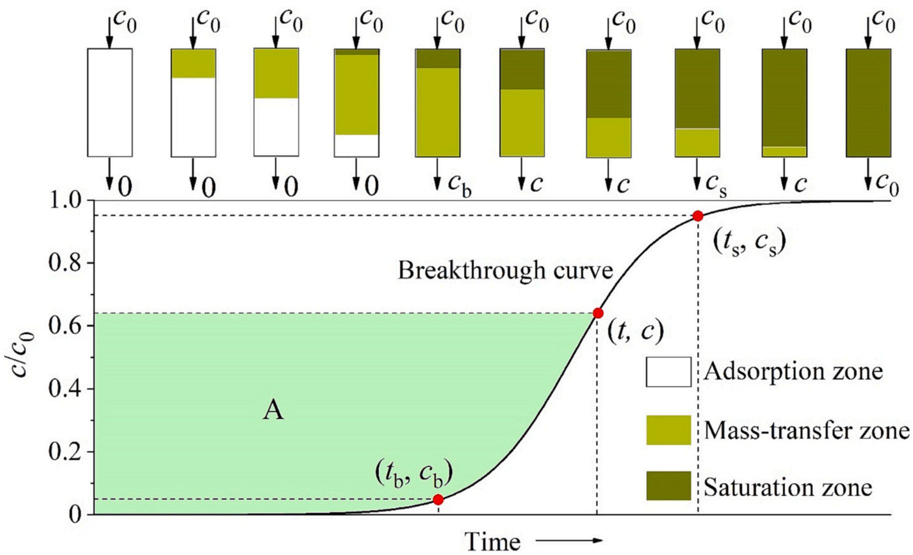

Fig. 1. Schematic diagram of MTZ traveling through a fixed-bed column and the breakthrough curve.

MTZ, in which the adsorbent particles accumulate the adsorbate molecules continuously from the feed and the amount adsorbed in the bed increases from zero to q_0. The shape and length of the MTZ largely depend on the rate of adsorption and the shape of the isotherm curve [33]. During the adsorption process, the MTZ travels through the bed with a velocity that is much slower than the influent velocity. The breakthrough occurs for the first time when the MTZ reaches the end of the bed. The operating time required is called the breakthrough time (t_b). For a given flow rate, the better adsorbable the contaminant is, the later the breakthrough occurs. In practice, the breakthrough time is often defined as the operating time required to reach the minimum detectable or maximum allowable concentration of the contaminant to be removed [34]. As time goes on, the MTZ gradually moves forward, while the saturation zone expands. Also, the concentration of the contaminant rapidly rises at the outlet. The saturation is reached when the entire MTZ leaves the column. At this moment, no net adsorption takes place in the bed and the related time is known as the saturation time (t_s).

The profile of the relative concentration of a contaminant at the outlet over time is called the breakthrough curve. The S-shaped breakthrough curve is a consequence of the decrease in the driving force for mass transfer from the fluid to the solid phase [35]. The breakthrough curve is a mirror of the MTZ and also affected by the rate of adsorption and the shape of the isotherm curve. To be specific, it is primarily influenced by the operating parameters such as flow rate, influent concentration, bed height, pH and particle size [36,37]. When the shape of the breakthrough curve is as sharp as possible, the most effective adsorption performance can be obtained [38]. The position of the breakthrough curve at the t-axis depends on the moving velocity of the MTZ, which in turn depends on the flow velocity and the strength of adsorption. Given that the asymptotic properties of the breakthrough models, they cannot exactly predict the breakthrough time and the saturation time. The operating times at $c/c_0 = 0.05$ and 0.95 are popularly defined as the breakthrough time and the saturation time, respectively [39].

A phenomenological model for adsorption of water contaminants is developed according to the following assumptions [40–42]: (i) isothermal condition; (ii) plug flow in the axial direction; (iii) negligible radial dispersion; (iv) linear driving force for the solid adsorption; (v) uniformly spherical adsorbent particles; and (vi) constant geometric dimensions, interstitial velocity and void fraction. The mass balance equation in the liquid phase is expressed as [43].

$$\frac{\partial c}{\partial t} + u\frac{\partial c}{\partial z} + \frac{1-\varepsilon}{\varepsilon}\frac{\partial q_t}{\partial t} = D_L\frac{\partial^2 c}{\partial z^2} \tag{1}$$

The axial dispersion coefficient is estimated by the following eq. [44].

$$D_L = ud_p\left(\frac{20}{\varepsilon}\frac{D_m}{ud_p} + \frac{1}{2}\right) \tag{2}$$

The molecular diffusivity D_m can be calculated by the Stokes–Einstein eq. [45]. A linear driving force model for the solid kinetics is written as [46].

$$\frac{\partial q_t}{\partial t} = k_s(q_e - q_t) \tag{3}$$

The mass-transfer processes in the solid phase include film diffusion, intraparticle diffusion and adsorption on the active sites [10]. The equilibrium data is analyzed by the Langmuir model, which is expressed as

$$q_e = \frac{q_{max}K_Lc_e}{1 + K_Lc_e} \tag{4}$$

The initial and boundary conditions are given as

$$t = 0; c = 0\ q = 0 \tag{5}$$

$$t = 0; z = 0\ c = c_0 \tag{6}$$

$$t = 0; z = 0\ \frac{\partial c}{\partial z} = \frac{u}{D_L}(c_e - c) \tag{7}$$

$$t = 0; z = x\ \frac{\partial c}{\partial z} = 0 \tag{8}$$

The accurate estimation of the equilibrium loading is a precondition for calculating other process parameters such as the removal efficiency and length of the MTZ [47]. The equilibrium loading can be obtained by integration of the measured breakthrough curve. As shown in Fig. 1, the amount adsorbed at any time is numerically proportional to the area A enclosed by the breakthrough curve, the vertical axis and the straight line $c/c_0 = 1$, which is expressed as [48].

$$q_t = \frac{vc_0A}{1000m} = \frac{vc_0}{1000m}\int_0^t\left(1 - \frac{c}{c_0}\right)dt \tag{9}$$

The breakthrough and saturation capacities can be obtained when the upper limits of integral are t_b and t_s, respectively. Other vital process parameters can refer to Supplementary material.

3. Breakthrough models

The appropriate design of a fixed-bed adsorber requires the development of a sound mathematical model to describe the dynamic adsorption behaviors and predict the breakthrough curves [30]. Eq. (1) is more general and mathematically rigid, but it requires complicated numerical solutions. The breakthrough models with the simple analytical solutions receive increasing attention because they can provide the sufficient fitting accuracy without the need for complex numerical solutions [9]. However, the model parameters involved are empirical in a sense, which are extremely sensitive to the process parameters such as flow rate, initial solute concentration, adsorbent particle size, pH and temperature. Thus, the practicality of these lumped parameters that embed some operating features is limited to the range of the operating conditions investigated [49]. In view of the complexity of the phenomenological models, the use of simpler and more tractable models appears more suitable and logical. For practical purposes, the selection of an optimal model should obtain a compromise between the accuracy of the process description and the effort for determining the model parameters [50].

3.1. Traditional breakthrough models

3.1.1. Bohart–Adams model

The Bohart–Adams model is initially used to describe the adsorption of chlorine on the charcoal in a fixed-bed column. The adsorption process is primarily controlled by the available active sites on the adsorbent surface [51]. It provides a high fitting quality under different operating conditions. The Bohart–Adams model assumes that the adsorption reaction is not instantaneous but proportional to the residual adsorption capacity of the adsorbent and the concentration of the target contaminant to be treated in the bulk solution, which is expressed as [14].

$$\frac{c}{c_0} = \frac{exp(k_{BA}c_0t)}{exp\left(\frac{k_{BA}a_0x}{u}\right) + exp(k_{BA}c_0t) - 1} \tag{10}$$

The Bohart–Adams model has been successfully extended to description of the adsorption of water contaminants [52,53]. In the denominator of Eq. (10), the last term is entirely negligible except for very small values of the first two exponential terms [54]. Thus, the Bohart–Adams model can reduce to

$$\frac{c}{c_0} = \frac{1}{1 + exp\left[k_{BA}c_0\left(\frac{a_0x}{uc_0} - t\right)\right]} \tag{11}$$

To facilely account for the rationality of this simplification, Eq. (10) is rewritten as [55].

$$\frac{c}{c_0} = \frac{1}{1 + exp\left[k_{BA}c_0\left(\frac{1}{k_{BA}c_0}ln\left[exp\left(\frac{k_{BA}a_0x}{u}\right) - 1\right] - t\right)\right]} \tag{12}$$

Both $\frac{1}{k_{BA}c_0}ln\left[exp\left(\frac{k_{BA}a_0x}{u}\right) - 1\right]$ and $\frac{a_0x}{uc_0}$ correspond to the operating time required to reach half of the influent concentration ($c/c_0 = 0.5$). One can readily see that Eq. (11) and Eq. (12) are mathematically equivalent and thereby provide the same fitted curves and error statistics. But, the differences between the model parameters (k_{BA} and a_0) in Eq. (11) and Eq. (12) are still uncertain. To address this confusion, according to the L'Hôpital's rule [56], the following limit is acceptable.

$$\lim_{k_{BA}\to\infty}\frac{1}{k_{BA}c_0}ln\left[exp\left(\frac{k_{BA}a_0x}{u}\right) - 1\right] = \frac{a_0x}{uc_0} \tag{13}$$

It is obvious that the term $\frac{1}{k_{BA}c_0}ln\left[exp\left(\frac{k_{BA}a_0x}{u}\right) - 1\right]$ increases with the increase in k_{BA} and infinitely approaches $\frac{a_0x}{uc_0}$. In general, it is desirable to have larger rate of adsorption (k_{BA}) and higher capacity of the adsorbent (a_0) for a given continuous system. This ensures that the difference between $\frac{a_0x}{uc_0}$ and $\frac{1}{k_{BA}c_0}ln\left[exp\left(\frac{k_{BA}a_0x}{u}\right) - 1\right]$ is considerably small. As a result, the simplification of Eq. (10) to Eq. (11) is rational. On the other hand, it is debatable to regard the linear form of Eq. (11) as the bed depth service time (BDST) model proposed by Hutchins [57] in many publications. By comparison, it is found that the BDST model is only a rearranged form of the Bohart–Adams model. Thus, Eq. (11) and its linear form should not be called the BDST model in the future studies.

3.1.2. Thomas model

The Thomas model is one of the most widely used breakthrough models, which can be used to estimate the rate constant and saturation capacity for given operating conditions [58]. The derivation of the Thomas model is based on the following assumptions [59,60]: (i) The plug flow occurs in the bed with no axial dispersion; (ii) The rate of adsorption is determined by chemical effects rather than diffusion; and (iii) The adsorption process obeys the Langmuir adsorption kinetics. It is theoretically appropriate to evaluate the adsorption process with extremely small external and internal diffusion resistances [61]. The Thomas model has a sound theoretical basis, which is expressed as [15].

$$\frac{c}{c_0} = \frac{1}{1 + exp\left[k_Tc_0\left(\frac{q_0m}{\nu c_0} - t\right)\right]} \tag{14}$$

In practice, the Thomas model is widely used for modeling of the breakthrough curves regardless of linear or nonlinear isotherm involved in a fixed-bed adsorption system [18]. Eq. (14) and Eq. (11) have the same mathematical form. The dimensions of k_T and k_{BA} are consistent, representing the second-order reaction rate constants. Let $\frac{q_0m}{\nu c_0} = \frac{a_0x}{uc_0}$, then $a_0 = \frac{q_0m}{V_{bed}}$ (V_{bed}, volume of bed). Thus, the physical meaning of the parameter a_0 is the weight of solute uptake per unit volume of bed not solution. It should be emphasized that it is unrealistic to assume reaction kinetics as the only rate-determining step [62]. Nowadays, the mass transfer-based models largely supersede reaction-based models because the former is more practical. Even so, the Bohart–Adams and Thomas models are still preferred by many researchers.

3.1.3. Yoon–Nelson model

The Yoon–Nelson model can accurately predict the breakthrough curves in the entire range of time and the service life of the bed. It has not only a simple mathematical form but also requires no detailed information on the characteristics of water contaminants of interest, type of the adsorbent and physical properties of the bed [63]. The Yoon–Nelson model assumes that the rate of decrease in the probability of adsorption for each molecule is proportional to the probability of adsorption and the probability of breakthrough, which is expressed as [16].

$$\frac{c}{c_0} = \frac{1}{1 + exp[k_{YN}(\tau - t)]} \tag{15}$$

The rationale for this assumption is based on the observation that the rate of change in the breakthrough concentration at a specific time is proportional to the breakthrough concentration at the outlet and the number of the active sites of the adsorbent particles in the bed [16]. According to the previous study [64], the Yoon–Nelson model is mathematically a logistic function, in which k_{YN} determines the degree of curvature of the breakthrough curve, while τ determines its location at the t-axis. Compared with the Bohart–Adams and Thomas models, the determination of the parameters k_{YN} and τ does not need to consider any operating conditions, which makes the curve fitting more convenient. In addition, Yoon and Nelson further examine the effect of the influent concentrations by introducing a new parameter a while keeping other conditions unchanged. For a given value of c/c_0, the effects of two influent concentrations on the operating time and the weight of solute adsorbed (w_t, mg) are expressed as [65].

$$\frac{t_1}{t_2} = \left(\frac{c_{0,2}}{c_{0,1}}\right)^a; \frac{w_{t,2}}{w_{t,1}} = \left(\frac{c_{0,2}}{c_{0,1}}\right)^{1-a} \tag{16}$$

Note that Eq. (16) also provides a practical means of predicting the values of t_b and τ as well as w_e for various influent concentrations.

3.1.4. Clark model

The development of the Clark model is based on the use of a mass-transfer concept in combination with the Freundlich isotherm [17]. The Clark model is initially employed to examine the adsorption performance of granular activated carbon for removal of low concentrations of organic compounds. Its underlying assumptions include that [48,66]: (i) Ideal plug flow occurs in a fixed-bed column; (ii) The film diffusion is the rate-controlling step; (iii) All solutes are removed at the outlet of column; and (iv) The shape of the MTZ keeps unchanged. The analytical solution of the Clark model is expressed as [17].

$$\frac{c}{c_0} = \frac{1}{[1 + A \cdot exp(-rt)]^{\frac{1}{n-1}}} \tag{17}$$

where

$$A = \left(\frac{{c_0}^{n-1}}{{c_b}^{n-1}} - 1\right) \cdot exp(rt_b) \tag{18}$$

$$r = \frac{K_T \mu}{u}(n-1) \tag{19}$$

The Freundlich isotherm in the Clark model is written as [67].

$$q_e = K_F {c_e}^{\frac{1}{n}} \tag{20}$$

Strictly speaking, the values of the parameters K_F and n are not determined by batch experiments at equilibrium but column experiments at saturation. That is, the saturation adsorption capacities q_0 are first obtained at different influent concentrations c_0, and then the Freundlich model is used to fit the experimental data (a plot of q_0 versus c_0) by the nonlinear regression. However, it is highly unreasonable to use the parameter n determined by batch experiments to analyze the breakthrough data in some recent studies [68,69]. In our opinion, since the Freundlich model is embedded in the Clark model, the parameter n can be seen as an undetermined parameter during the curve fitting, so that the fitting ability of the three-parameter Clark model will be significantly improved. To facilely identify the curve characteristics of the Clark model, Eq. (17) is rewritten as.

$$\frac{c}{c_0} = \frac{1}{\left\{1 + exp\left[r\left(\frac{1}{r}lnA - t\right)\right]\right\}^{\frac{1}{n-1}}} \tag{21}$$

It can be clearly seen that Eq. (21) and Eq. (15) have similar mathematical forms and the Clark and Yoon–Nelson models are equivalent at $n = 2$. Namely, the Bohart–Adams, Thomas and Yoon–Nelson models can be regarded as special cases of the Clark model. Since n is an adjustable parameter, the fitted curve provided by the Clark model is always asymmetric except for $n = 2$. It is predicted that the Clark model is superior to the Bohart–Adams, Thomas and Yoon–Nelson models in terms of the fitting accuracy. Moreover, an obvious drawback of these four models is that their relative concentration c/c_0 is not equal to zero at $t = 0$, probably resulting in a poor fit for some data points in the initial stage of the breakthrough curves.

3.1.5. Wolborska model

The Wolborska model is originally used to describe p-nitrophenol adsorption on activated carbon. Its underlying assumptions include that [46,70]: (i) The low-concentration region is formed in a residence time; (ii) The initial concentration distribution moves along the column at a constant velocity; (iii) The width of the breakthrough curve is constant in the low-concentration region; (iv) The low-concentration region is characterized by the constant kinetic coefficients; and (v) The rate of adsorption is controlled by the external mass transfer (i.e. film diffusion and axial diffusion). The Wolborska model is derived by the mass balance equation (Eq. (1)) without considering void fraction of bed, which is expressed as [71].

$$\frac{c}{c_0} = exp\left(\frac{\beta_a c_0}{a_0}t - \frac{\beta_a}{u}x\right) \tag{22}$$

where

$$\beta_a = \frac{(u-\mu)^2}{2D_L}\left[\sqrt{1 + \frac{4\beta_0 D_L}{(u-\mu)^2}} - 1\right] \tag{23}$$

The effective kinetic coefficient β_a reflects the effect of both axial dispersion and external mass transfer [72]. The axial diffusion is negligible for a low bed height or high flow rate, i.e. $\beta_a = \beta_0$ [73]. The first term for the exponential term in Eq. (22) is not dimensionless because it possesses a ratio of volumes of bed to solution [18]. To address this problem, Wolborska and Pustelnik further propose the following eq. [74].

$$\frac{c}{c_0} = exp\left(\frac{\beta_a c_0 \varepsilon}{\rho q_0}t - \frac{\beta_a}{u}x\right) \tag{24}$$

Unfortunately, the correct form of the Wolborska model still attracts little attention [70,75]. It is worth noting that Eq. (22) and Eq. (24) are mathematically an exponential function, which do not match the shape of the complete breakthrough curve in the fixed-bed column [64]. Consequently, the characteristic parameters β_a and a_0 (or q_0) do not objectively reflect the performance of a fixed-bed column. The frequent use of the Wolborska model to describe the breakthrough curves is caused by the fact that its mathematical characteristics are ignored. In practice, the Wolborska model works only at low concentrations. The fitting ability of the Wolborska model for modeling of the complete breakthrough curve is highly questioned. Thus, it is not recommended to continue using the Wolborska model in the future studies.

3.1.6. Modified dose-response model

The Bohart–Adams, Thomas and Yoon–Nelson models have two disadvantages: A symmetric breakthrough curve and a poor fit for initial data points. Thus, a comparatively larger deviation may exist between the fitted curves and the experimental data. Yan et al. develop the modified dose-response model based on statistical analysis of the experimental data [35]. This model can minimize the errors resulting from the use of the Bohart–Adams, Thomas and Yoon–Nelson models, especially at lower or higher time periods of the breakthrough curve [76], which is written as

$$\frac{c}{c_0} = 1 - \frac{1}{1 + \left(\frac{V}{b}\right)^a} \tag{25}$$

The empirical parameter a determines the slope of the fitted curve, while b represents the breakthrough volume reaching half a maximum response, i.e. $b = V$ at $c/c_0 = 0.5$. The modified dose-response model can provide an asymmetric S-shaped curve at $a > 1$ [64]. Its fitting quality is often higher than the Bohart–Adams, Thomas and Yoon–Nelson models. Yan et al. further find that the corresponding operating times are $\frac{b}{\nu}$ (by definition of $V = \nu t$) and $\frac{q_0 m}{\nu c_0}$ respectively at $c/c_0 = 0.5$ for the modified dose-response and Thomas models. Assuming that $\frac{b}{\nu} = \frac{q_0 m}{\nu c_0}$ (i.e. $b = \frac{q_0 m}{c_0}$), the modified dose-response model is rewritten as [35].

$$\frac{c}{c_0} = 1 - \frac{1}{1 + \left(\frac{\nu c_0}{q_0 m}t\right)^a} \tag{26}$$

Since the Bohart–Adams, Thomas and Yoon–Nelson models are mathematically equivalent (see below), the parameter b also meets $b = \frac{\nu a_0 x}{u c_0} = \nu\tau$. This approach seems reasonable, but it is actually controversial. As shown in Fig. 2, two function curves of the modified dose-response and Thomas models intersect at points (b/v, 0.5) or ($\frac{q_0 m}{\nu c_0}$, 0.5). The area enclosed by three curves (respective function curves, the vertical axis and the straight line $c/c_0 = 0.5$) is not equal due to $A_2 \neq A_3$. The intersection of their function curves does not necessarily mean that their parameters can be interchanged. Thus, Eq. (26) is not recommended to fit the experimental data. The saturation capacity for the modified dose-response model can be easily calculated by Eq. (9). Recently, Lee et al. systematically compare the fitting performance of the above six models and recommend the Bohart–Adams, Clark and modified dose-response models for modeling of the breakthrough curves [77].

3.1.7. Klinkenberg model

It is very desirable to find a quick method to determine the model parameters with a satisfactory accuracy. For practical purposes, extreme accuracy in the solution of Eq. (1) is never required. In a fixed-bed column, the exact solution of the equations describing transient mass-transfer phenomena under given operating conditions can be approximated by an error function [78]. The Klinkenberg model assumes that [79]: (i) The one-dimensional flow occurs in a fixed-bed column; (ii) There is no axial diffusion in the direction of flow; and (iii) The rate of mass transfer is proportional to the concentration gradient at the solid/solution interface. An approximate analytical solution derived by Klinkenberg is expressed as [78].

$$\frac{c}{c_0} = \frac{1}{2}\left[1 + erf\left(\sqrt{\tau} - \sqrt{\zeta} + \frac{1}{8\sqrt{\tau}} + \frac{1}{8\sqrt{\zeta}}\right)\right] \tag{27}$$

where

$$\zeta = \frac{K_f a x}{u} \tag{28}$$

$$\tau = \frac{K_f a K}{(1-\varepsilon)}\left(t - \frac{\varepsilon x}{u}\right) \tag{29}$$

The error function or probability integral erf(x) is expressed as.

$$erf(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-u^2} du \tag{30}$$

A comparison of exact and approximate values shows that the relative concentration c/c_0 has a maximum error of ±0.6 % ($\zeta = 2$), ±0.2 % ($\zeta = 4$) and ± 0.1 % ($\zeta = 8$) for $\tau \geq 1$. The error approaches zero with the further increase in ζ. It is not recommended to use the Klinkenberg model for $\zeta < 2$ and $\tau < 1$. According to Eq. (29), the Klinkenberg model is only applied to the cases where $t > \varepsilon x/u$, may result in a lack of some data points in the initial stage of the breakthrough curves. Moreover, this work also corrects the improper expressions of the dimensionless coefficients ζ and τ reported in the literature. Since the error function is an odd function, the Klinkenberg model can provide a symmetric breakthrough curve. It has been widely used to describe adsorption of arsenic [80], hydrogen sulfide [81] and guaiacol [82]. To our knowledge, the Bohart–Adams, Thomas and Yoon–Nelson models are mathematically a logistic function, while the Klinkenberg model is the error function. It is expected that two types of the models should have similar fitting ability because of the symmetric properties. Their main difference consists in the degree of curvature, which can be reduced by adjusting the model parameters. Recently, Dima et al. develop a simper model to analyze Cr(VI) adsorption on chitosan flakes based on the error function, characterizing the breakthrough curve by two parameters under the assumption of a normal distribution: linear velocity of movement of the adsorption zone v_d and standard deviation of the position of the adsorption zone front σ_a, which is expressed as [83].

$$\frac{c}{c_0} = \frac{1}{2}\left[1 + erf\left(\frac{v_d t - x}{\sqrt{2}\,\sigma_a}\right)\right] \tag{31}$$

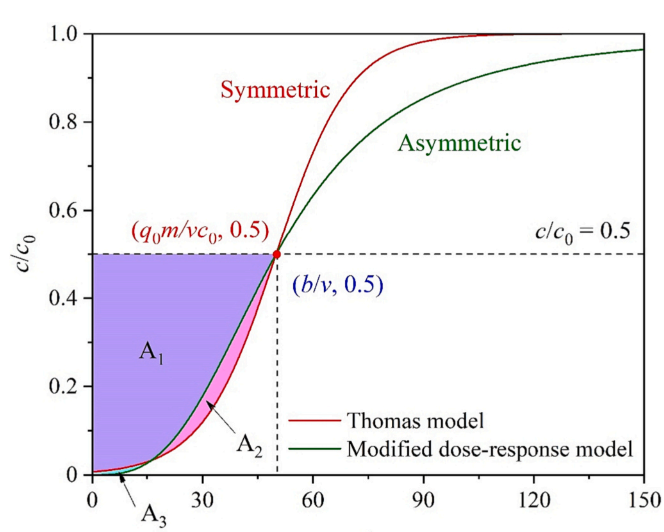

Fig. 2. Schematic diagram for comparison of Thomas and modified dose-response models.

3.1.8. Chern–Chien model

Based on the constant pattern concept of the wave propagation theory, Chern and Chien develop a mathematical model to successfully predict the breakthrough curves of p-nitrophenol adsorption on granular activated carbon [84]. The Chern–Chien model assumes that [85]: (i) No chemical reactions occur in the packed column; (ii) Only mass transfer by convection is significant; (iii) Radial and axial dispersions are negligible; (iv) The flow pattern is the ideal plug flow; (v) The temperature in the column is uniform and invariant with time; (vi) The flow rate is constant and invariant with the column position; and (vii) The rate of adsorption is described by the linear driving force. When an adsorption process follows the Langmuir isotherm, the Chern–Chien model is expressed as [84].

$$t = t_{1/2} + \frac{\rho q_0}{\varepsilon K_f a c_0}\left[ln2x + \frac{1}{1 + K_L c_0} ln\frac{1}{2(1-x)}\right] \tag{32}$$

Pan et al. derive another form of the Chern–Chien model when an adsorption process obeys the Freundlich isotherm, which is expressed as [86].

$$t = t_{1/2} + \frac{\rho q_0}{\varepsilon K_f a c_0}\left[ln2x - \frac{1}{n-1} ln\frac{1 - x^{n-1}}{1 - 2^{1-n}}\right] \tag{33}$$

The volumetric mass-transfer coefficient ($K_f a$) is considered as an undetermined parameter during the curve fitting. The Chern–Chien model gives an excellent fit for prediction of the breakthrough curves in recent studies [87–89]. However, the fitting results may deviate from the real situation when the rate of adsorption is controlled by the intraparticle diffusion. According to Eq. (32) and Eq. (33), the relative concentration must meet $0 < c/c_0 < 1$ since the value of the logarithmic term must be more than zero, meaning that some data points are left out in the initial and final stages of the breakthrough curves during the curve fitting. According to symmetry of function [90], it can be easily proved that the function curves of Eq. (32) and Eq. (33) are symmetric if and only if $K_L = 1/c_0$ and $n = 2$. Thus, the Chern–Chien model represents an asymmetric S-shaped curve apart from the above special case.

This work attempts to solve the Chern–Chien model (Langmuir-type and Freundlich-type) by OriginPro software. The experimental data required are extracted from adsorption of Ni(II) [91] and methyl orange

[92] by Engauge Digitizer 12.1 software. The Chern–Chien model is mathematically an implicit function and thus its curve fitting needs to adopt the iteration algorithm of orthogonal distance regression (ODR). In the iterative process, the ODR algorithm minimizes the sum of squares of the orthogonal distances from the observed data to the fitted curve rather than the deviations between the observed and predicted values for the dependent variable [93]. Therefore, the error statistics do not objectively reflect the goodness of fit of the Chern–Chien model (see Table S1). As shown in Fig. 3, the fitted curves provided by two types of the Chern–Chien model agree well with the experimental data. On the whole, the Freundlich-type Chern–Chien model is superior to Langmuir-type Chern–Chien model. This may be ascribed to the fact that the Freundlich isotherm has the ability to describe the multilayer adsorption on the heterogeneous surface [67]. It is observed from Table 1 that process variables can significantly influence values of the model parameters ($K_{\mathrm{f}}a$ and $t_{1/2}$).

Last but not least, the difference in the adsorption parameters exists in the batch and fixed-bed operations [94,95]. In a batch reactor, the mass-transfer driving force or rate of adsorption usually decreases due to the continuous decrease in concentration during the adsorption process. By contrast, the adsorbent particles are always in contact with the inlet concentration in a fixed-bed adsorber, resulting in a high driving force over the whole process [33]. For this reason, the parameters obtained in batch experiments are not applicable to the fixed-bed system. The parameters K_{L} and n in the Chern–Chien model should be determined by the measured breakthrough data at saturation in the fixed-bed system. Similar to the Clark model, the Chern–Chien model may be more applicable if regarding K_{L} and n as the unknown parameters during the curve fitting.

3.2. *Fractal-like breakthrough models*

The classical rate equations for description of the dynamic evolution of chemical processes contain time-independent rate constants and transport properties [24]. Its implicit assumption is that the process evolves in a well-stirred system with homogenous spatial distribution of reactants. As a consequence, the classical reaction kinetics is not applicable for the diffusion-limited heterogeneous processes, which are found to be unsatisfactory when the reactants are spatially constrained by either walls, phase boundaries or force fields on the microscopic level [96]. The adsorbent particles are of high heterogeneity in a fixed-bed column, which stems from the physical heterogeneity produced by the surface with fractal-like geometries (i.e. porous adsorbents) and the chemical heterogeneity caused by different functional groups present on the adsorbent surfaces (different adsorption sites) [97]. A distinguishing feature of heterogeneous reactions in fractal-like geometries is that the

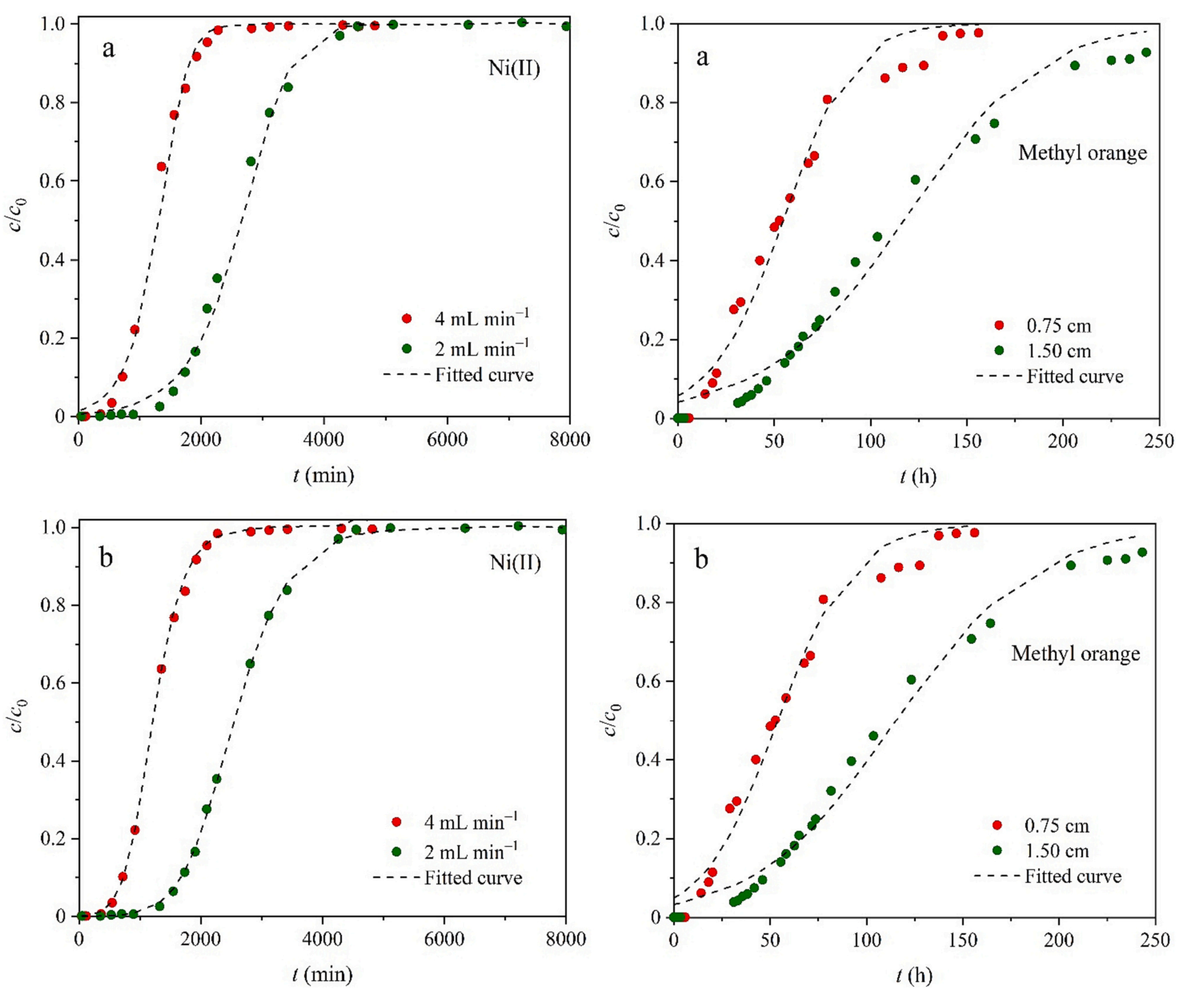

Fig. 3. Breakthrough curves for adsorption of Ni(II) and methyl orange: (a) Langmuir-type Chern-Chien model, (b) Freundlich-type Chern-Chien model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Fitting results of the Chern-Chien model for adsorption of Ni(II) and methyl orange.

Contaminant	ϕ (cm)	m (g)	x (cm)	v (mL min^{-1})	c_0 (meq L^{-1})	ρ (g L^{-1})	q_0 (meq g^{-1})	ε	K_L (L meq^{-1})	n	Langmuir-type		Freundlich-type	
											$K_f a$ (min^{-1})	$t_{1/2}$ (min)	$K_f a$ (min^{-1})	$t_{1/2}$ (min)
Ni(II)	2.8	8	30.5	2	2.14	41.359	1.468	0.875	0.46	1.46	0.057	2676	0.132	2537
				4	2.11	41.361	1.350	0.870			0.089	1317	0.222	1210

Contaminant	ϕ (mm)	m (g)	x (cm)	v (mL h^{-1})	c_0 (mg L^{-1})	ρ (g L^{-1})	q_0 (mg g^{-1})	ε	K_L (L mg^{-1})	n	Langmuir-type		Freundlich-type	
											$K_f a$ (h^{-1})	$t_{1/2}$ (h)	$K_f a$ (h^{-1})	$t_{1/2}$ (h)
Methyl orange	10	0.15	0.75	4.5	15	455	20.2	0.335	0.018	1.864	84.3	54.6	100.4	54.0
		0.30	1.50		15		21.6				47.1	117.2	57.0	115.3

rate coefficient decreases with time [98]. The interplay of energetic and geometric heterogeneities results in the fractal-like kinetics, which provides new insights into the adsorption phenomena at the solid/solution interface. According to the fractal-like kinetic theory, the rate constant k of the classical reaction kinetics can be expressed by a time-dependent rate constant, which is given as [96].

$$k = k_0 t^{-h} \ (t \geq 1, 0 \leq h \leq 1) \tag{34}$$

Here, the parameter h is a heterogeneity parameter related to the spectral dimension of fractal systems [99]. Eq. (34) is valid for diffusion in homogeneous systems when $h = 0$. Compared with traditional breakthrough models that focus only on the decrease in the driving force, the fractal-like breakthrough models reflect the slowdown of the process because the value of k decreases with time as the power law. Haerifar and Azizian suppose that temporal variation of the rate coefficient (fractal-like kinetics) during the adsorption process can be explained by either the occurrence of available slower diffusion pathways for homogeneous surfaces or the progressive occupation of active sites characterized by greater activation energies for heterogeneous surfaces [100,101]. The first explanation implies that the rate coefficient of desorption from a flat surface is time-independent [102]. The fractal-like kinetics allows a more realistic understanding of the mechanisms controlling the process evolution on a microscopic scale, which in turn is a prerequisite for the reliable design of large-scale reactors [103]. In the fixed-bed systems, the dynamic evolution of the adsorption space and diffusion-limited heterogeneous processes create fundamental conditions for the applicability of the fractal-like kinetics. The Bohart–Adams, Thomas, Yoon–Nelson and Clark models are derived by constant rate coefficient or mass-transfer coefficient, may lead to a poor fit in some cases. The recent studies apply the fractal-like concept to these models, leading to [1,97,104].

$$\frac{c}{c_0} = \frac{exp\left(\frac{k_{\mathrm{BA},0} c_0}{1-h} t^{1-h}\right)}{exp\left(\frac{k_{\mathrm{BA},0} t^{-h} a_0 x}{u}\right) + exp\left(\frac{k_{\mathrm{BA},0} c_0}{1-h} t^{1-h}\right) - 1} \tag{35}$$

$$\frac{c}{c_0} = \frac{1}{1 + exp\left(\frac{k_{\mathrm{T},0} t^{-h} q_0 m}{\nu} - \frac{k_{\mathrm{T},0} c_0}{1-h} t^{1-h}\right)} \tag{36}$$

$$\frac{c}{c_0} = \frac{1}{1 + exp\left[\frac{k_{\mathrm{YN},0}}{1-h}\left(\tau^{1-h} - t^{1-h}\right)\right]} \tag{37}$$

$$\frac{\mathrm{c}}{\mathrm{c}_0} = \frac{1}{\left[1 + \mathrm{A}_0 \cdot exp\left(-\frac{\mathrm{r}}{1-\mathrm{h}} \mathrm{t}^{1-\mathrm{h}}\right)\right]^{\frac{1}{\mathrm{n}-1}}} \tag{38}$$

The previous study has proved that the sum of the first two exponential terms in the denominator of Eq. (35) is much >1 [104]. Thus, the fractal-like Bohart–Adams can reduce to.

$$\frac{c}{c_0} = \frac{1}{1 + exp\left(\frac{k_{\mathrm{BA}} t^{-h} a_0 x}{u} - \frac{k_{\mathrm{BA},0} c_0}{1-h} t^{1-h}\right)} \tag{39}$$

The above fractal-like breakthrough models can reduce to the corresponding Bohart–Adams, Thomas, Yoon–Nelson and Clark models at $h = 0$. It is worth noting that Eq. (37) and Eq. (38) are mathematically equivalent through simple transformations and other three fractal-like breakthrough models also equivalent. The previous study reported that the Bohart–Adams, Thomas and Yoon–Nelson models represented the symmetric S-shaped curves [64], which deviate from the measured breakthrough curves in most cases. By contrast, the fractal-like Bohart–Adams, fractal-like Thomas and fractal-like Yoon–Nelson models have the ability to describe the asymmetric breakthrough curve at $h \neq 0$. Thus, three fractal-like breakthrough models are more applicable. Secondly, three fractal-like breakthrough models can better describe the heterogeneity and complexity of the adsorbents because they take into account the fractal structure of porous adsorbents. Thirdly, three fractal-like breakthrough models consider the distribution characteristics of permeability in porous adsorbents, which are capable of describing the differences in permeability inside the adsorbent and is conducive to predicting the mass-transfer steps more accurately. Finally, the introduction of the hydrodynamic properties and fractal characteristics of porous adsorbents makes three fractal-like breakthrough models allow for a more comprehensive consideration of the effect of different factors on the mass-transfer processes. In summary, the fractal-like kinetics provides a more realistic theoretical basis for the understanding of the microscopic mechanisms dominating the process evolution, which is in turn a prerequisite for the reliable design of large-scale fixed-bed adsorbers.

3.3. Recent empirical breakthrough models

As mentioned above, the equilibrium loading for the single-solute adsorption in the fixed-bed column is proportional to the area enclosed by the vertical axis, the breakthrough curve and the straight line $c/c_0 = 1$. Also, the breakthrough time and the saturation time can be easily obtained if a model is capable of predicting the breakthrough curves well. Hence, these results provide a realistic basis for establishment of the empirical breakthrough models. From a mathematical perspective, a complete breakthrough curve can be divided into two types: symmetric and asymmetric. Most often, it is asymmetric even for the single-solute adsorption [105]. Blagojev et al. observe the asymmetric breakthrough curves for Cu(II) adsorption on sugar beet shreds, which are caused by the presence of two types of active sites on the adsorbent surface [106]. Most likely, adsorption of Cu(II) on two types of active sites occurs simultaneously, but the change in their dominance appears with time in the whole adsorption process. Based on the two-stage adsorption mechanism, the parallel sigmoidal model proposed in their study is expressed as.

$$\frac{c}{c_0} = p\left[1 - \frac{1}{1+\left(\frac{t}{\tau_1}\right)^{k_1}}\right] + (1-p)\left[1 - \frac{1}{1+\left(\frac{t}{\tau_2}\right)^{k_2}}\right] \tag{40}$$

The parameter p reflects the proportion of each part in two-stage adsorption mechanism. The parallel sigmoidal model is represented by the superposition of two equations analogous to the modified dose-response model. Eq. (40) can reduce to the modified dose-response model when one of two-stage adsorption mechanisms controls the process evolution ($p = 0$ or 1). The roles of the parameters k_i and τ_i ($i = 1,2$) are similar to that of a and b in the modified dose-response model. The parallel sigmoidal model is also used to describe Cr(VI) adsorption on three agricultural wastes (sugar beet shreds, poplar sawdust and wheat straw) and shows perfect fitting performance under different operating conditions [107]. Recently, some S-shaped functions are adopted to establish the empirical breakthrough models by introducing two unknown parameters. Hu et al. first propose three empirical breakthrough models based on the logistic, hyperbolic tangent and double exponential (Gompertz) functions, which are expressed as [108].

$$\frac{c}{c_0} = \frac{1}{1+exp[k(\tau - t)]} \tag{41}$$

$$\frac{c}{c_0} = \frac{1}{2}\cdot\{1+tanh[k(t-\tau)]\} \tag{42}$$

$$\frac{c}{c_0} = exp\{-exp[k(\tau - t)]\} \tag{43}$$

Later, Chu also proposes an empirical breakthrough model based on the Gompertz function, which is expressed as [109].

$$\frac{c}{c_0} = exp[-exp(\alpha - \beta t)] \tag{44}$$

It is evident that Eq. (43) and Eq. (44) are similar in mathematical forms, but Chu does not explain the physical meanings of the empirical parameters α and β. It is interesting to find that logistic (Eq. (41)) and Yoon–Nelson (Eq. (15)) models are mathematically equivalent. Thus, the parameters k and τ for the logistic, hyperbolic tangent and Gompertz models can be also seen as the lumped parameters that determine the degree of curvature of the breakthrough curve and its location at the t-axis, respectively. In order to sufficiently express the subtle change in the breakthrough curve, Hu et al. further define four characteristic parameters i.e. maximum specific breakthrough rate μ_{max}, lag time λ, inflection point t_i and half-operating time t_{50} [108]. In order to concisely introduce this work, the modified breakthrough models can refer to our previous studies [64,108] based on the parameters μ_{max} and λ. In addition, Hu et al. further propose two empirical breakthrough models based on error and Gudermannian functions, which are expressed as [29].

$$\frac{c}{c_0} = \frac{1}{2}\cdot\{1+erf[k(t-\tau)]\} \tag{45}$$

$$\frac{c}{c_0} = \frac{1}{2}\cdot\left\{1+\frac{2}{\pi}\cdot arctan(sinh[k(t-\tau)])\right\} \tag{46}$$

It should be emphasized that the above empirical breakthrough models are obtained through appropriate transformation of the corresponding mathematical functions. As a result, the Gompertz model represents an asymmetric S-shaped curve. By contrast, the logistic, hyperbolic tangent, error and Gudermannian models represent a symmetric S-shaped curve and their main difference lies in degree of curvature. In order to equip these four models with the ability to describe the asymmetric penetration curve, the concept of fractal-kike kinetics can be introduced. The physical meanings of k and τ for these models are the rate constant and the time required to reach the inflection point, respectively [29,108].

The Weibull distribution function is initially proposed to measure the probability of failure of some technical indexes [110]. It has a simple mathematical expression and satisfies the necessary general characteristics of the breakthrough curve. Chu establishes an empirical breakthrough model based on the appropriate form of the Weibull distribution function, which is given as [111].

$$\frac{c}{c_0} = 1 - exp\left[-\left(\frac{t}{\tau}\right)^k\right] \tag{47}$$

A distinct advantage of the Weibull model for modeling of the breakthrough curves is that the relative concentration $c/c_0 = 0$ at $t = 0$, which can provide a higher fitting accuracy for description of the breakthrough curves in the initial stage. Here, we analyze the curve characteristics of the Weibull model. As shown in Fig. 4a, all curves pass through one point (τ, 1–1/e) when the parameter k is adjusted. The shape of the curve represented by the Weibull model depends on the value of k. A convergent L-shaped curve occurs when $0 < k \leq 1$. By contrast, an asymmetric S-shaped curve appears at $k > 1$ and its degree of curvature will become larger with the increase in the value of k. As shown in Fig. 4b, the value of the parameter τ can influence both degree of curvature and shape of the S-shaped curve and it will become more precipitous with the decrease in the value of τ. The adjustable parameters k and τ in Weibull model make the S-shaped curve more diverse. It is expected that Eq. (47) is capable of describing the dynamic adsorption behaviors well in the fixed-bed column. To explore the breakthrough rate, the first-order derivative of Eq. (47) is expressed as.

$$\frac{d(c_t/c_0)}{dt} = \frac{k\cdot t^{(k-1)}}{\tau^k}\cdot exp\left[-\left(\frac{t}{\tau}\right)^k\right] \tag{48}$$

As shown in Fig. 4c and Fig. 4d, the breakthrough rate profiles resemble the bell-shaped curve. The parameters k and τ for the Weibull model jointly determine the position and amplitude of the rate profiles. The rate profiles are also asymmetric and have an asymptote of $\frac{d(c_t/c_0)}{dt} = 0$ at $t \to \infty$. These results will contribute to gaining insights into the rate evolution in a fixed-bed column. The detail interpretation with respect to the rate profiles can refer to our previous study [1].

Given that the Bohart–Adams, Thomas and Yoon–Nelson models give a poor fit for modeling of the asymmetric breakthrough curve, Apiratikul and Chu modify these three models by a logarithmic transformation to improve their fitting quality, which are expressed as [112].

$$\frac{c}{c_0} = \frac{1}{1+exp\left[k_{BA}ln\left(\frac{a_0 x}{u}\right) - k_{BA}ln(c_0 t)\right]} \tag{49}$$

$$\frac{c}{c_0} = \frac{1}{1+exp\left[k_T ln\left(\frac{q_0 m}{\nu}\right) - k_T ln(c_0 t)\right]} \tag{50}$$

$$\frac{c}{c_0} = \frac{1}{1+exp[k_{YN}ln(\tau) - k_{YN}ln(t)]} \tag{51}$$

The above three modified models are capable of more accurately predicting the breakthrough curves and providing reliable estimates for the breakthrough time and the saturation time. These new models only contain two undetermined parameters that appear in the original models, providing an alternative strategy for modeling of the asymmetric breakthrough curve due to the presence of the logarithmic term that contains time t. Based on this idea, Chu and Hashim further propose the log-Gompertz model [5] and the log-normal distribution [113], which is expressed as.

$$\frac{c}{c_0} = exp[-exp(\alpha - \beta ln(t))] \tag{52}$$

$$\frac{c}{c_0} = \frac{1}{2}\cdot\left\{1+erf\left[\frac{ln(t)-a}{\sqrt{2}\,b}\right]\right\} \tag{53}$$

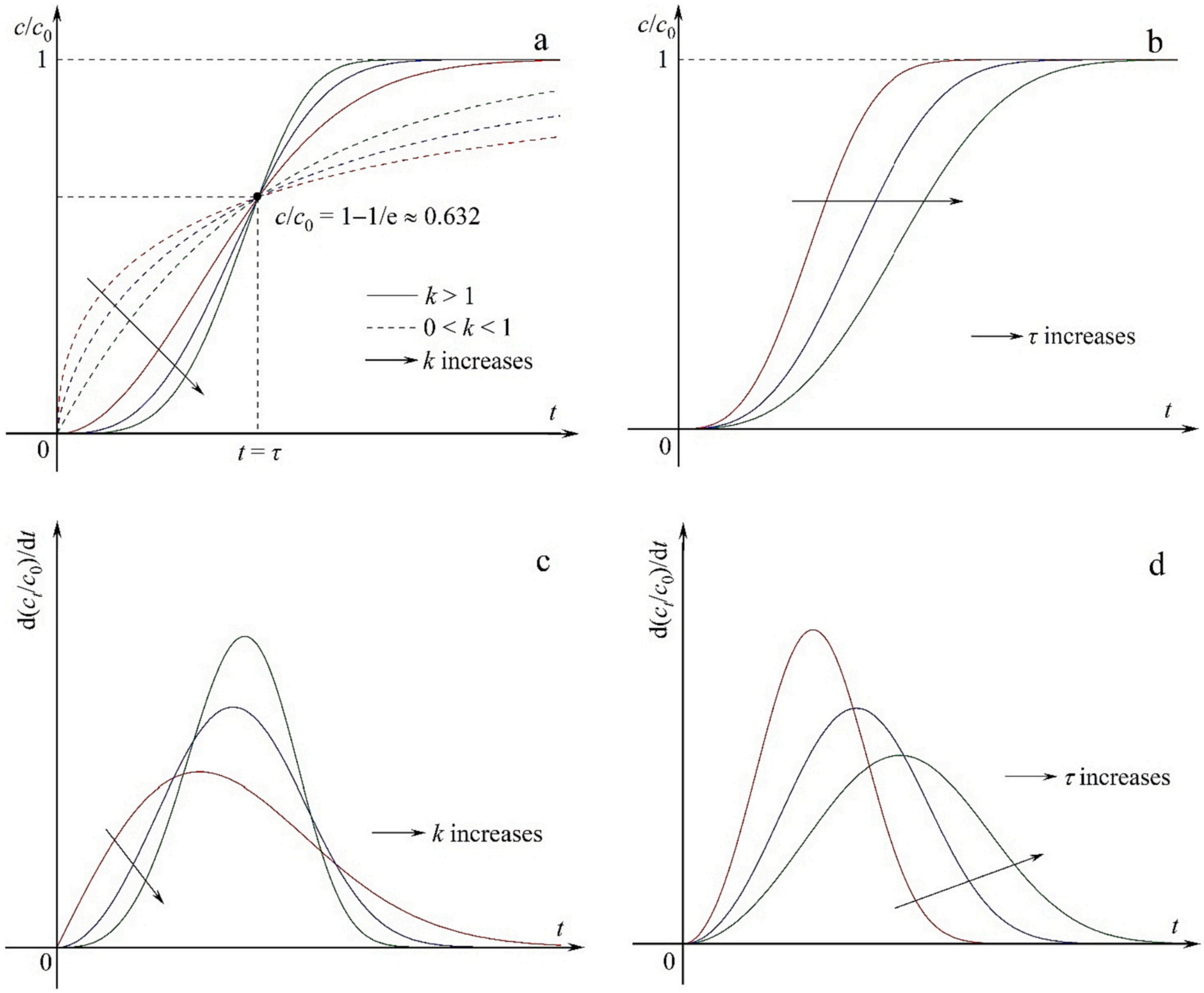

Fig. 4. Curve characteristics of the Weibull model and its first-order derivative.

It should be noted that the logarithmic terms that contain time t in Eqs. (49)–(53) keep dimensionless by introducing a set of variables/parameters with a value of unity artificially. This modification is meaningless in practice. In fact, the reason why Eqs. (49)–(53) have the ability to describe the asymmetric breakthrough curve is because they change the mathematical structure of the original models. To some extent, Eqs. (49)–(51) lose their theoretical basis and the model parameters also lack the physical meanings. From a mathematical perspective, the value of all logarithmic terms must be more than zero, i.e. $t > 1/c_0$ Eq. (49) and Eq. (50) or $t > 1$ for Eqs. (51)–(53). In this case, the fitted curve may deviate from the breakthrough curves in the initial stage of adsorption.

Recently, Singh et al. propose an empirical breakthrough model by a combination of the Avrami equation with the breakthrough curves characteristics, which is expressed as [114].

$$\frac{c}{c_0} = 1 - exp(-kt^n) \tag{54}$$

One of its advantages is that the relative concentration c/c_0 is also equal to zero at $t = 0$. As shown in Fig. S2, the parameters k and n can influence the degree of curvature and position of the curves simultaneously. The value of n determines the type of the function curve, representing an asymmetric S-shaped curve at $n > 1$ and a L-shaped curve at $0 < n \leq 1$. Two adjustable parameters k and n make the curve fitting more flexible. Thus, Eq. (54) may be expected to provide the high fitting accuracy for description of many breakthrough curves.

4. Comprehensive evaluation of fitting quality

Although there are many mechanistic and empirical mathematical models available for modeling of the breakthrough curves in a fixed-bed column, their fitting quality is not sufficiently evaluated. In most fixed-bed studies, the selection of the breakthrough models seems considerably arbitrary when correlated with the experimental data. Recently, Kimani compare the fitting performance of some breakthrough models systematically based on the measured breakthrough data from bisphenol A adsorption on polyaniline [115]. In general, a good fit requires that the distribution of the experimental data should match the curve characteristics of the breakthrough models well [43]. The fitting quality of the breakthrough models can be visually evaluated by how close the fitted curve is to the data points. If the curve fitting succeeds, a plot of the predicted versus observed values will produce a straight line that evenly passes through the data points [104]. In addition, error statistics are also used to quantitatively assess the fitting quality of the breakthrough models. The coefficient of determination (R^2) is a good measure of the goodness of fit, which is expressed as [25].

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \acute{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y}_i)^2} \tag{55}$$

where n is the number of the data points, y_i is the observed value, $\acute{y}_i$ is the predicted value, $\bar{y}_i$ is the mean value of all observed data.

The value of R^2 close to 1 indicates that the fit is often a good one

[116]. However, a larger value of R^2 does not necessarily mean a better fit since the degrees of freedom can influence R^2. The value of R^2 will rise if more parameters are added to the models, but this does not imply a better fit [67]. In addition, R^2 is sensitive to extreme data points and also influenced by the range of the independent variable [117]. The use of R^2 is particularly inappropriate if the models are obtained by different transformations of the response scale [118]. Sole reliance on R^2 may fail to reveal important data characteristics and model inadequacies. Thus, R^2 is not an adequate criterion for evaluation of the fitting quality. The adjusted R^2 (Adj. R^2) overcomes these disadvantages and may be a better metric, which is expressed as [119].

$$\mathrm{Adj.}R^2 = 1 - \left(1 - R^2\right)\left(\frac{n-1}{n-p}\right) \tag{56}$$

Obviously, Adj. R^2 overcomes the problem of the rise in R^2, especially for the dataset with a small sample size. Other frequently used error statistics are summarized in Table S2. However, it is not sufficient enough to evaluate the fitting quality by error statistics alone. The residual plot is a more reliable evaluation criterion than error statistics, which is recommended to further diagnose the fitting results [120]. It can be used to examine the underlying statistical assumptions (i.e. constant variance, independence of variables and normality of the distribution) about residuals and provide valuable information on how to improve the models. If the fitting quality is acceptable, all residuals should tend to fall in a horizontal band centered around zero, showing no systematic tendencies toward a clear pattern [28]. Statistically speaking, rather than asking whether a particular fitting result is good, it is more appropriate to compare two fitting results. Current studies lack further analysis of the fitting results. In this work, the *F*-test and Akaike's information criterion (AIC) are used to compare the fitting results from two models for the same dataset, which are expressed as [121,122].

$$F = \frac{(\mathrm{RSS}_1 - \mathrm{RSS}_2)/(\mathrm{df}_1 - \mathrm{df}_2)}{\mathrm{RSS}_2/\mathrm{df}_2} \tag{57}$$

$$\mathrm{AIC} = \begin{cases} \mathrm{n}\ln\left(\frac{\mathrm{RSS}}{\mathrm{n}}\right) + 2\mathrm{p} \quad \left(\frac{\mathrm{n}}{\mathrm{p}} \geq 40\right) \\ \mathrm{n}\ln\left(\frac{\mathrm{RSS}}{\mathrm{n}}\right) + 2\mathrm{p} + \frac{2\mathrm{p}(\mathrm{p}+1)}{\mathrm{n}-\mathrm{p}-1} \quad \left(\frac{\mathrm{n}}{\mathrm{p}} < 40\right) \end{cases} \tag{58}$$

where RRS denotes the residual sum of square, df is the degree of freedom, the subscripts 1 and 2 correspond to the simple and complex models, respectively.

The *F*-test and AIC are available to determine which model is the better and thereby improve the predictive and explanatory abilities of the model. The *F*-test assumes that two models are nested, where one model is a simplified version of the other [123]. It takes advantage of the difference in RSS of each fit to find out which model is the best. The *F*-test is commonly used in linear regression or analysis of variance, which can examine the significant differences between two nested models. The significance of *F*-test consists in determining whether the regression coefficients in the model are significantly different and whether the model itself is significant. The cumulative distribution function of the *F*-distribution is assessed to yield a *p* value. If the *p* value is statistically significant ($p < 0.05$), the fitting ability of the complex model is better than the simpler model [122]. Otherwise, the complex model is rejected. The *F*-test does not decide which model is correct and provides information on the goodness of fit [123]. The objective of AIC is to find out a model that can best interpret the experimental data but contain the fewest model parameters. AIC can provide robust and precise estimates based on maximum likelihood to rank the models rather than concept of significance [124]. AIC has a larger penalty for overfitting and thus helps to avoid selecting overly complex models. The significance of AIC is to obtain a compromise of the complexity of the estimated models and their goodness of fit. Compared with the *F*-test, AIC can compare nested or non-nested models. Hence, any two models can be compared by AIC and the one with a smaller value of AIC is suggested to be optimal. The Akaike's weight (W_A) indicates the probability of a better model and can be used to further confirm the optimal model, which can be expressed as [125].

$$W_A = \frac{1}{1 + exp(0.5\Delta\mathrm{AIC})} \tag{59}$$

where ΔAIC is the difference between the AIC values of one model and the best model.

Taking the Bohart–Adams and fractal-like Bohart–Adams models as an example, this work aims to compare the fitting results of ciprofloxacin [126] and phenol and *p*-nitrophenol [86] adsorption by using OriginPro software. The *F*-test and AIC can be obtained by the following procedures. Step 1: perform the curve fitting twice by the Bohart–Adams and fractal-like Bohart–Adams models for the same dataset and create two fitting reports automatically. Step 2: select **Analysis → Fitting → Compare Model** from the Origin menu, and then open **Fitting: fitcmpmodel** dialog box. Step 3: specify the way to recalculate and update the result, fitting results, comparison method, fit parameters, fit statistics, and click **OK** (see Fig. S1). The results of model comparison will show in the **Book** dialog box. As shown in Fig. 5, the breakthrough curve of ciprofloxacin adsorption shows an asymmetric distribution. Compared with the Bohart–Adams model, the fitted curve provided by the fractal-like Bohart–Adams model is more consistent with the experimental data. Its predicted values are very close to the observed values. The corresponding residuals fluctuate randomly in the vicinity of zero and fall in a narrow horizontal band. It is observed from Table 2 that the values of RSS, reduced χ^2 and RMSE are smaller and that R^2 and Adj. R^2 are larger for the fractal-like Bohart–Adams model. The *p* value for *F*-test is <0.05, indicating that this comparison is significant. The value of AIC is also smaller and its weight almost approaches unity for the fractal-like Bohart–Adams model. Consequently, ciprofloxacin adsorption follows the fractal-like Bohart–Adams model. By contrast, the breakthrough curves of phenol and *p*-nitrophenol adsorption show a symmetric distribution. Thus, the Bohart–Adams and fractal-like Bohart–Adams models have comparable fitting abilities and the fitted curves are almost coincident. The fractal-like Bohart–Adams model is superior to the Bohart–Adams model in terms of various error statistics for phenol adsorption. It is worth noting that all error values were fairly close for *p*-nitrophenol adsorption. In this case, error statistics are not sufficient enough to evaluate the fitting quality. The *F*-test and AIC are particularly important for further diagnosis of the fitting results. One can readily see that the *p* value for *F*-test is much more than 0.05. The Bohart–Adams model has smaller value of AIC and larger value of W_A. This case indicates that a model with more parameters does not necessarily mean a better fit.

5. Existing problems

5.1. Data quality

High quality of the experimental data is a prerequisite for obtaining a good fit. In many cases, the data quality is purposely or unconsciously neglected when the theoretical models are used to analyze the breakthrough curves. The poor data quality is not conducive to the accurate prediction of the model parameters and insights into the dynamic adsorption behaviors in a fixed-bed column. To our knowledge, the relevant situations mainly include that: (i) Many studies lack the indispensable repeated experiments (no error bars); (ii) The measured breakthrough curves are not complete (not S-shaped) [48]; (iii) The data points fluctuate greatly [127]; and (iv) The breakthrough curves intersect under different experimental conditions [128]. In our opinion, the first two cases can be easily solved by the standardized operation procedures. The reason why the latter two cases exist is that the hydration process (volume expansion) occurs when the adsorbent is in contact with the influent or the dominant mass-transfer and reaction mechanisms change during the adsorption process.

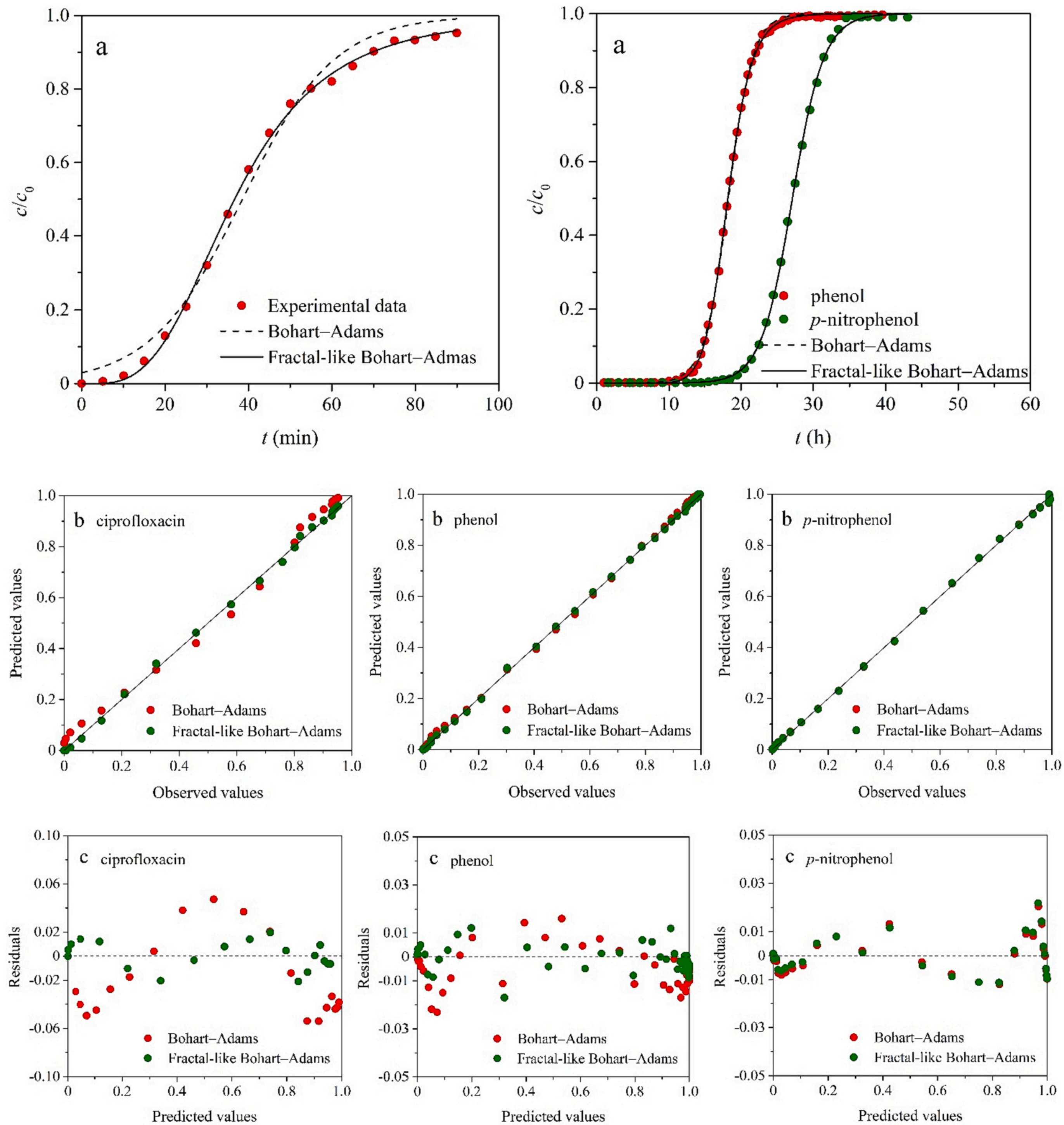

Fig. 5. Comparison of Bohart–Adams and fractal-like Bohart–Adams models: (a) fitted curves, (b) predicted versus observed values and (c) residual plot.

5.2. Partial and complete breakthrough curves

Since it is often a tedious and time-consuming work for obtaining the complete breakthrough curve, some studies focus more on how many bed volumes are required for breakthrough to occur [129]. Although the partial breakthrough curve can be obtained in a relatively short time, the completeness of the experimental data can directly influence the model parameters to be estimated. Besides, the partial breakthrough curve does not profoundly reveal the dynamic adsorption behaviors in a fixed-bed column. Taking the Thomas model (Eq. (14)) as an example, this work examines the fitting results of the partial and complete breakthrough curves from adsorption of methylene blue [37]. As shown in Fig. 6, the fitted curves provided by the Thomas model agree well with the partial and complete breakthrough curves at different influent concentrations and an error statistic Adj. R^2 is >0.99 in all cases. The fitting results of other partial breakthrough curves refer to Fig. S3. However, there are significant differences in k_T and q_0 obtained from the partial and complete breakthrough curves. As shown in Table 3, compared with the complete breakthrough curves, the maximum relative errors of k_T are 110.8 %, 26.1 % and 47.5 % respectively at 50, 100 and 150 mg L^{-1}, while the corresponding maximum relative errors of q_0 are −20.5 %, −4.0 % and −13.4 %. In general, the curve fitting aims to purely

Table 2
Fitting results of ciprofloxacin, phenol and *p*-nitrophenol adsorption with Bohart–Adams and fractal-like Bohart–Adams models.

Parameters and error statistics	Ciprofloxacin[a]		Parameters and error statistics	Phenol[b]		Parameters and error statistics	*p*-Nitrophenol[b]	
	Bohart–Adams	Fractal-like Bohart–Adams		Bohart–Adams	Fractal-like Bohart–Adams		Bohart–Adams	Fractal-like Bohart–Adams
k_{BA} (L mg min^{-1})	4.03×10^{-4}	–	k_{BA} (L mmol h^{-1})	0.113	–	k_{BA} (L mmol h^{-1})	8.64×10^{-2}	–
$k_{BA,0}$ (L mg min^{h-1})	–	1.17×10^{-3}	$k_{BA,0}$ (L mmol h^{h-1})	–	0.222	$k_{BA,0}$ (L mmol h^{h-1})	–	9.85×10^{-2}
a_0 (mg L^{-1})	669	1245	a_0 (mmol L^{-1})	594	1023	a_0 (mmol L^{-1})	882	936
h	–	0.487	h	–	0.422	h	–	5.82×10^{-2}
RRS	2.76×10^{-2}	2.47×10^{-3}	RRS	4.54×10^{-3}	1.52×10^{-3}	RRS	1.75×10^{-3}	1.77×10^{-3}
Reduced χ^2	1.62×10^{-3}	1.55×10^{-4}	Reduced χ^2	8.56×10^{-5}	2.92×10^{-5}	Reduced χ^2	5.16×10^{-5}	5.38×10^{-5}
R^2	0.9890	0.9990	R^2	0.9995	0.9998	R^2	0.9997	0.9997
Adj. R^2	0.9883	0.9989	Adj. R^2	0.9995	0.9998	Adj. R^2	0.9997	0.9997
RMSE	4.03×10^{-2}	1.24×10^{-2}	RMSE	9.25×10^{-3}	5.41×10^{-3}	RMSE	7.18×10^{-3}	7.33×10^{-3}
F-test	162.5 ($p = 8.55 \times 10^{-10} < 0.05$)		F-test	103.2 ($p = 6.03 \times 10^{-14} < 0.05$)		F-test	0.413 ($p = 0.525 > 0.05$)	
AIC	−116.6	−159.1	AIC	−510.7	−568.5	AIC	−350.7	−347.7
W_A	5.71×10^{-10}	1.000	W_A	2.80×10^{-13}	1.000	W_A	0.817	0.183

[a] operating conditions of $c_0 = 225$ mg L^{-1}, $u = 1.91$ mL min, $x = 25$ cm.
[b] operating conditions of $c_0 = 5.32$ mmol L^{-1}, $u = 34.9$ mL h^{-1}, $x = 5.71$ cm.

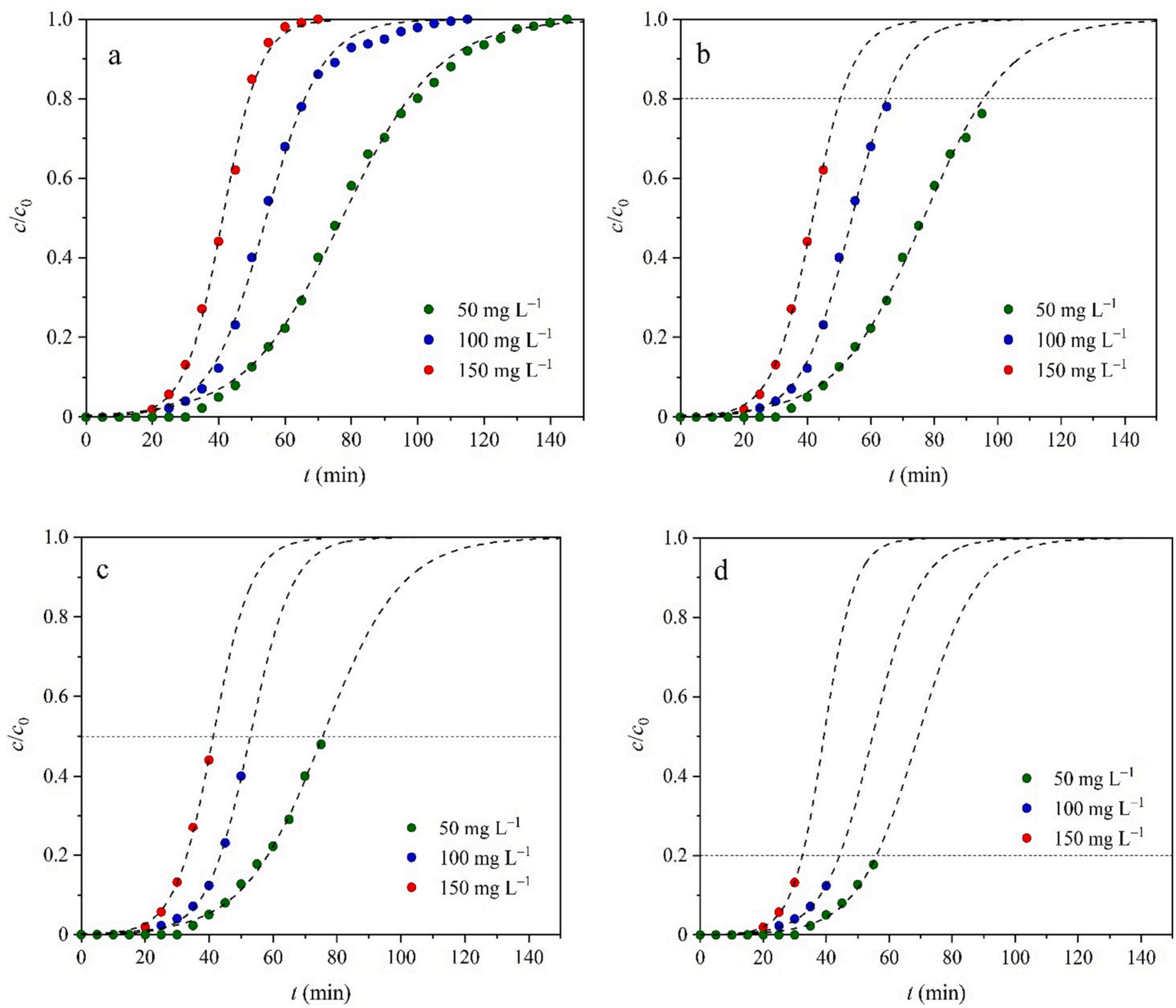

Fig. 6. Complete and partial breakthrough curves of methylene blue adsorption with the Thomas model: (a) 100 %, (b) 80 %, (c) 50 % and (d) 20 %. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3
Fitting results for complete and partial breakthrough curves of methylene blue adsorption at different concentrations.

c/c_0	50 mg L^{-1}			100 mg L^{-1}			150 mg L^{-1}		
	k_T (L mg^{-1} min^{-1})	q_0 (mg g^{-1})	Adj. R^2	k_T (L mg^{-1} min^{-1})	q_0 (mg g^{-1})	Adj. R^2	k_T (L mg^{-1} min^{-1})	q_0 (mg g^{-1})	Adj. R^2
1.0	1.39×10^{-3}	52.20	0.9977	1.19×10^{-3}	73.42	0.9978	1.18×10^{-3}	83.55	0.9984
0.9	1.40×10^{-3}	52.16	0.9957	1.23×10^{-3}	73.28	0.9975	1.14×10^{-3}	83.73	0.9966
0.8	1.48×10^{-3}	51.80	0.9959	1.29×10^{-3}	73.02	0.9977	1.06×10^{-3}	84.42	0.9980
0.7	1.57×10^{-3}	51.28	0.9970	1.33×10^{-3}	72.72	0.9978	1.06×10^{-3}	84.42	0.9980
0.6	1.60×10^{-3}	51.12	0.9961	1.40×10^{-3}	72.25	0.9978	1.12×10^{-3}	83.64	0.9974
0.5	1.62×10^{-3}	51.00	0.9941	1.50×10^{-3}	71.37	0.9979	1.12×10^{-3}	83.64	0.9974
0.4	1.67×10^{-3}	50.61	0.9914	1.50×10^{-3}	71.37	0.9979	1.24×10^{-3}	81.67	0.9978
0.3	1.66×10^{-3}	50.70	0.9842	1.40×10^{-3}	72.55	0.9958	1.24×10^{-3}	81.67	0.9978
0.2	2.11×10^{-3}	46.70	0.9826	1.33×10^{-3}	73.90	0.9872	1.35×10^{-3}	79.57	0.9933
0.1	2.93×10^{-3}	41.48	0.9654	1.48×10^{-3}	70.50	0.9681	1.74×10^{-3}	72.36	0.9962

minimize the deviations between the fitted curves and the data points when a mathematical model is used to analyze the breakthrough curves, ignoring the physical meanings of the model parameters. As a consequence, the model parameters obtained from the partial breakthrough curves are not applied to the design and optimization of the fixed-bed adsorber. On the whole, the complete breakthrough curves are recommended to obtain the model parameters.

5.3. Oversimplification of Bohart–Adams model

In addition to two forms of the Bohart–Adams model mentioned above, an oversimplified form of the Bohart–Adams model frequently appears in the recent literature, which is expressed as [130].

$$\frac{c}{c_0} = exp\left(k_{BA}c_0t - \frac{k_{BA}a_0x}{u}\right) \tag{60}$$

The simplification of Eq. (11) to Eq. (60) is completely unreasonable. From a mathematical point of view, Eq. (60) is an exponential function rather than a logistic function through oversimplification of the Bohart-Adams model, which covers up the functionality of the original Bohart-Adams model and greatly weakens the ability to describe the breakthrough curve [131]. By comparing Eq. (22), Eq. (24) and Eq. (60), it is found that the Wolborska and oversimplified Bohart–Adams models are equivalent in mathematical nature, i.e. $k_{BA} = \frac{\beta_a}{a_0}$ or $\frac{\beta_a \varepsilon}{\rho q_0}$. The fitted curve provided by Eq. (60) seriously deviates from the complete breakthrough curve, especially at larger values of t. For this reason, the fitting ability of Eq. (60) is also increasingly questioned. A poor fit is bound to result in the inaccurate estimation of the model parameters, which is detrimental to understanding the dynamic adsorption behaviors and optimizing the design of the adsorber [54]. In our opinion, the exponential term $exp\left[k_{BA}c_0\left(\frac{a_0x}{uc_0} - t\right)\right]$ in Eq. (11) contains the independent valuable t, whose value decreases with the increase in t. This exponential term will be much <1 when the value of t is sufficiently large. In this case, the unity term is not neglected in the denominator of Eq. (11). Hence, the use of Eq. (60) should be avoided in the future studies. In order to generalize the Bohart–Adams model and enable it to describe the asymmetric breakthrough curve, Hu et al. derive the n-order Bohart–Adams model, which is expressed as [104].

$$\frac{c}{c_0} = \left[1 + na_0^{1-n}c_0^{n-1}\left(\left[\frac{1 + (n-1)\frac{k_na_0c_0^{n-1}x}{u}}{1 + (n-1)k_na_0^{n-1}c_0t}\right]^{\frac{1}{n-1}} - \left[\frac{1}{1 + (n-1)k_na_0^{n-1}c_0t}\right]^{\frac{1}{n-1}}\right)\right]^{-\frac{1}{n}} \tag{61}$$

5.4. Equivalent models

In some recent papers [132,133], the Bohart–Adams, Thomas and Yoon–Nelson models show different fitting results (fitted curves and error values) when used to analyze the breakthrough curves. In other words, these three models are assumed to be separate and independent. To our knowledge, A lack of the understanding of their intrinsic mathematical relationships can account for the further propagation of such a mistake. The previous study has demonstrated that these three models were mathematically a logistic function that represents a symmetric S-shaped curve and their parameters are interchangeable, which are expressed as [1].

$$k_{YN} = k_{BA}c_0 = k_Tc_0 \tag{62}$$

$$\tau = \frac{a_0x}{uc_0} = \frac{q_0m}{\nu c_0} \tag{63}$$

Hence, the Bohart–Adams, Thomas and Yoon–Nelson models are equivalent. Their fitted curves should be coincident and all error values are equal when the curve fitting is carried out for the same set of the experimental data. The above relationships reveal the absurd behaviors of treating these three models as independent models and comparing their respective fitting abilities based on various error statistics. According to Eq. (62), the parameters k_{BA} and k_T are regarded as the second-order reaction rate constants. Similar to the parameter τ, the physical meanings of the terms $\frac{a_0x}{uc_0}$ and $\frac{q_0m}{\nu c_0}$ represent the operating time required to reach 50 % breakthrough [64]. According to Eq. (62) and Eq. (63), the determination of k_{BA}, k_T, a_0 and q_0 avoids the repeated curve fitting. The revelation of intrinsic relationships between the Bohart–Adams, Thomas and Yoon–Nelson models will contribute to precisely obtaining the model parameters and better understanding the dynamic adsorption behaviors in a fixed-bed column. Just as the roles of k_{YN} and τ, the terms $k_{BA}c_0$ (k_Tc_0) and $\frac{a_0x}{uc_0}$ $\left(\frac{q_0m}{\nu c_0}\right)$ determine the degree of curvature of the breakthrough curve and its location at the t-axis, respectively. The values of k_{YN} and τ depend on the operating conditions and to some extent may be regarded as the lumped parameters that embed some physical processes and operating features. From a mathematical perspective, the same breakthrough curve is obtained regardless of the process parameters only if the value of the terms $\frac{a_0x}{uc_0}$ and $\frac{q_0m}{\nu c_0}$ keeps constant.

5.5. Linear and nonlinear curve fitting

The accurate calculation of the model parameters plays an important role in evaluation the separation efficiency of the contaminant and identification of its mass-transfer law [134]. The values of the model parameters are affected by the regression method. Although it is highly questioned, the linear fitting is still frequently adopted by many researchers. The linearization of a model requires the corresponding transformation of the measured experimental data. The linearized treatment implicitly alters their error structure and may also violate the error variance and normality assumptions of standard least squares [135]. As a consequence, different linearized forms of a model may result in different estimates of the undetermined parameters [136]. In addition, the linearized treatment may cause arbitrary exclusion of certain data points from the estimated model for calculation as these data points are not in the domain of definition of the linearized equations. For instance, a linearized form of the Thomas model is expressed as.

$$ln\left(\frac{c_0}{c}-1\right)=\frac{k_T q_0 m}{\nu}-k_T c_0 t \quad (64)$$

It is universally acknowledged that the domain of definition of the logarithmic function is always more than zero and the bottom number of the power functions is not equal to zero. According to Eq. (64), one can get $0 < c < c_0$ or $0 < c/c_0 < 1$. When the Thomas model is used to fit the experimental data, the linearized treatment excludes data points at $c = 0$ and $c = c_0$. A new dependent variable $ln\left(\frac{c_0}{c}-1\right)$ is produced after linearization and its values may not be sufficiently accurate during the calculation process using the measured data. The linearization of a breakthrough model also results in the statistical bias. The error increases dramatically when the value of the logarithmic term in Eq. (64) is very close to its limit, especially for the breakthrough curve before breakthrough point and after saturation point [131]. Besides, the imprecisions of slope and intercept obtained from Eq. (64) may also be incorporated into the model parameters during the curve fitting. Last but not least, the linear fitting fails to examine more complex models that may be in better agreement with the experimental data. The nonlinear fitting is a versatile way to obtain the model parameters without any transformation [137]. It not only overcomes the existing disadvantages of the linear fitting, but also can solve the models with multiple parameters. The nonlinear fitting can provide a more robust and reliable parameter estimation than the linear fitting.

5.6. Reasons for asymmetric breakthrough curve

The breakthrough behaviors in a fixed-bed column are subject to the operating parameters, geometric dimensions of the column, types of the contaminants and physicochemical properties of the adsorbent. It is reported that the complete breakthrough curve is usually asymmetric S-shaped even for the single-solute adsorption [105]. When the solution containing a solute with a high affinity is fed into the fixed-bed column at high concentrations, the breakthrough curve first shows a sharp rise and then a slow approach to saturation. Such early breakthrough and tailing are caused by the slow surface diffusion [138]. If the intraparticle diffusion controls the adsorption kinetics, as it is often the case in practice, the breakthrough curve is asymmetric with a long tailing [33]. Moreover, an asymmetric breakthrough curve may be also attributed to the fact that the adsorbent contains two or more constituents of unequal reactivity or the rate of adsorption falls off more rapidly than the residual capacity of the adsorbent [14]. Thus, the measured breakthrough curves are asymmetric in most cases, which account for a relatively poor fit for the Bohart–Adams, Thomas and Yoon–Nelson models.

6. Concluding remarks

The mathematical modeling of the breakthrough curves is a practically indispensable link for the design and optimization of a fixed-bed reactor. However, the traditional breakthrough models are not applicable for the diffusion-limited heterogeneous processes due to their implicit assumptions of time-independent rate constants and transport properties. The fractal-like theory provides new insights into the dynamic adsorption behaviors and allows to understand the mechanisms controlling the process evolution deeply on a microscopic scale. Furthermore, based on the similarity of the breakthrough curves and some S-shaped function curves, the empirical breakthrough models developed have been an alternative strategy for modeling of the measured breakthrough curves. In most cases, the complete breakthrough curves are asymmetric S-shaped even for the single-solute adsorption. During the curve fitting, high quality of the experimental data is the first step to obtaining a good fit. The selection of an appropriate mathematical model for prediction of the breakthrough curves requires not only taking into account its underlying assumptions, curve characteristics and application scope, but also allowing for data quality, fitting methods, error statistics and residual plot. *F*-test and AIC can be used to compare the fitting results of two breakthrough models for the same dataset, but the former requires the two models must be nested. The breakthrough models with asymmetric properties usually have higher fitting quality. The partial breakthrough curves result in the inaccurate estimation of the model parameters. The Bohart–Adams, Thomas and Yoon–Nelson models are mathematically equivalent and their parameters are interchangeable. The Wolborska and oversimplified Bohart–Adams models are mathematically an exponential function, which are not applicable for prediction of the complete breakthrough curves. The fitting quality of the Chern–Chien model can be evaluated by the residual plot rather than error statistics since it adopts the iteration algorithm of ODR. This review is expected to help readers better understand and use the breakthrough models with simple analytical solutions and avoid the further propagation of some existing mistakes and inconsistencies in the future studies.

CRediT authorship contribution statement

Qili Hu: Writing – original draft, Validation, Methodology, Investigation, Conceptualization. **Xingyue Yang:** Validation, Formal analysis, Data curation. **Leyi Huang:** Validation, Software, Data curation. **Yixi Li:** Validation, Software, Formal analysis. **Liting Hao:** Validation, Funding acquisition, Data curation. **Qiuming Pei:** Software, Formal analysis, Data curation. **Xiangjun Pei:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was supported by National Key R&D Program of China (2023YFC3007103) and Key Research and Development Programme of Sichuan Province (No. 2021YFQ0066).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jwpe.2024.105065.

References

[1] Q. Hu, Y. Xie, C. Feng, Z. Zhang, Fractal-like kinetics of adsorption on heterogeneous surfaces in the fixed-bed column, Chem. Eng. J. 358 (2019) 1471–1478.
[2] S. Aguirre-Contreras, R. Leyva-Ramos, R. Ocampo-Pérez, C.G. Aguilar-Madera, J. V. Flores-Cano, N.A. Medellín-Castillo, Mathematical modeling of breakthrough curves for 8-hydroxyquinoline removal from fundamental equilibrium and adsorption rate studies, J. Water Process. Eng. 54 (2023) 103967.

[3] D. Juela, M. Vera, C. Cruzat, X. Alvarez, E. Vanegas, Mathematical modeling and numerical simulation of sulfamethoxazole adsorption onto sugarcane bagasse in a fixed-bed column, Chemosphere 280 (2021) 130687.
[4] R. Antonelli, G.R.P. Malpass, M.G.C. da Silva, M.G.A. Vieira, Fixed-bed adsorption of ciprofloxacin onto bentonite clay: characterization, mathematical modeling, and DFT-based calculations, Ind. Eng. Chem. Res. 60 (2021) 4030–4040.
[5] K.H. Chu, M.A. Hashim, Removal of antibiotics through fixed bed adsorption: comparison of different breakthrough curve models, J. Water Process. Eng. 56 (2023) 104512.
[6] K.N. Son, J.A. Weibel, J.C. Knox, S.V. Garimella, Limitations of the axially dispersed plug-flow model in predicting breakthrough in confined geometries, Ind. Eng. Chem. Res. 58 (2019) 3853–3866.
[7] A. Bringas, E. Bringas, R. Ibañez, M.F. San-Román, Fixed-bed columns mathematical modeling for selective nickel and copper recovery from industrial spent acids by chelating resins, Sep. Purif. Technol. 313 (2023) 123457.
[8] X. Lin, Q. Huang, G. Qi, S. Shi, L. Xiong, C. Huang, X. Chen, H. Li, X. Chen, Estimation of fixed-bed column parameters and mathematical modeling of breakthrough behaviors for adsorption of levulinic acid from aqueous solution using SY-01 resin, Sep. Purif. Technol. 174 (2017) 222–231.
[9] K.L. Tan, B.H. Hameed, Insight into the adsorption kinetics models for the removal of contaminants from aqueous solutions, J. Taiwan Inst. Chem. Eng. 74 (2017) 25–48.
[10] P.Y.R. Suzaki, M.T. Munaro, C.C. Triques, S.J. Kleinübing, M.R. Fagundes Klen, R. Bergamasco, L.M. de Matos Jorge, Phenomenological mathematical modeling of heavy metal biosorption in fixed-bed columns, Chem. Eng. J. 326 (2017) 389–400.
[11] D.S.P. Franco, J.L.S. Fagundes, J. Georgin, N.P.G. Salau, G.L. Dotto, A mass transfer study considering intraparticle diffusion and axial dispersion for fixed-bed adsorption of crystal violet on pecan pericarp (Carya illinoensis), Chem. Eng. J. 397 (2020) 125423.
[12] O. Hamdaoui, Removal of copper(II) from aqueous phase by Purolite C100-MB cation exchange resin in fixed bed columns: modeling, J. Hazard. Mater. 161 (2009) 737–746.
[13] T.G. Myers, A. Cabrera-Codony, A. Valverde, On the development of a consistent mathematical model for adsorption in a packed column (and why standard models fail), Int. J. Heat Mass Transf. 202 (2023) 123660.
[14] G.S. Bohart, E.Q. Adams, Some aspects of the behavior of charcoal with respect to chlorine, J. Am. Chem. Soc. 42 (1920) 523–544.
[15] H.C. Thomas, Chromatography: a problem in kinetics, Ann. NY Acad. Sci. 49 (1948) 161–182.
[16] Y.H. Yoon, J.H. Nelson, Application of gas adsorption kinetics I. A theoretical model for respirator cartridge service life, Am. Ind. Hyg. Assoc. J. 45 (1984) 509–516.
[17] R.M. Clark, Evaluating the cost and performance of field-scale granular activated carbon systems, Environ. Sci. Technol. 21 (1987) 573–580.
[18] K.H. Chu, Breakthrough curve analysis by simplistic models of fixed bed adsorption: In defense of the century-old Bohart-Adams model, Chem. Eng. J. 380 (2020) 122513.
[19] Asadullah, L. Kaewsichan, K. Techato, Z.N. Qaisrani, M.S. Chowdhury, M. Yilmaz, Elimination of selected heavy metals from aqueous solutions using biochar and bentonite composite monolith in a fixed-bed operation, J. Environ. Chem. Eng. 10 (2022) 106993.
[20] Z. Jiang, M. Chen, X. Lee, Q. Feng, N. Cheng, X. Zhang, S. Wang, B. Wang, Enhanced removal of sulfonamide antibiotics from water by phosphogypsum modified biochar composite, J. Environ. Sci. 130 (2023) 174–186.
[21] M. Ahmad, N.M.A. Lubis, M. Usama, J. Ahmad, M.I. Al-Wabel, H.A. Al-Swadi, M. I. Rafique, A.S.F. Al-Farraj, Scavenging microplastics and heavy metals from water using jujube waste-derived biochar in fixed-bed column trials, Environ. Pollut. 335 (2023) 122319.
[22] M.S. Podder, C.B. Majumder, Biological detoxification of As(III) and As(V) using immobilized bacterial cells in fixed-bed bio-column reactor: prediction of kinetic parameters, Groundw. Sustain. Dev. 6 (2018) 14–42.
[23] J. Abdi, H. Abedini, MOF-based polymeric nanocomposite beads as an efficient adsorbent for wastewater treatment in batch and continuous systems: modelling and experiment, Chem. Eng. J. 400 (2020) 125862.
[24] M. Balsamo, F. Montagnaro, Fractal-like Vermeulen kinetic equation for the description of diffusion-controlled adsorption dynamics, J. Phys. Chem. C 119 (2015) 8781–8785.
[25] F.M. de Souza, O.A.A. dos Santos, Assessment of fixed bed adsorption of 2,4-D herbicide onto modified bentonite clay, Water Air Soil Pollut. 233 (2022) 158.
[26] A. Satya, A. Harimawan, G. Sri Haryani, M.A.H. Johir, L.N. Nguyen, L.D. Nghiem, S. Vigneswaran, H.H. Ngo, T. Setiadi, Fixed-bed adsorption performance and empirical modelingof cadmium removal using adsorbent prepared from the cyanobacterium *Aphanothece sp* cultivar, Environ. Technol. Innov. 21 (2021) 101194.
[27] M. Jafari, M.R. Rahimi, A. Asfaram, M. Ghaedi, H. Javadian, Experimental design for the optimization of paraquat removal from aqueous media using a fixed-bed column packed with *Pinus Eldarica* stalks activated carbon, Chemosphere 291 (2022) 132670.
[28] M.A. Basunia, T. Abe, Adsorption isotherms of barley at low and high temperatures, J. Food Eng. 66 (2005) 129–136.
[29] Q. Hu, Q. Huang, D. Yang, H. Liu, Prediction of breakthrough curves in a fixed-bed column based on normalized Gudermannian and error functions, J. Mol. Liq. 323 (2021) 115061.
[30] M.S. Shafeeyan, W.M.A. Wan Daud, A. Shamiri, A review of mathematical modeling of fixed-bed columns for carbon dioxide adsorption, Chem. Eng. Res. Des. 92 (2014) 961–988.
[31] Y. Lin, T.A. Kurniawan, M. Zhu, T. Ouyang, R. Avtar, M.H. Dzarfan Othman, B. T. Mohammad, A.B. Albadarin, Removal of acetaminophen from synthetic wastewater in a fixed-bed column adsorption using low-cost coconut shell waste pretreated with NaOH, HNO_3, ozone, and/or chitosan, J. Environ. Manag. 226 (2018) 365–376.
[32] A. Thirunavukkarasu, R. Nithya, R. Sivashankar, Continuous fixed-bed biosorption process: a review, Chem. Eng. J. Adv. 8 (2021) 100188.
[33] E. Worch, Adsorption Technology in Water Treatment: Fundamentals, Walter de Gruyter GmbH & Co KG, Processes and Modeling, 2012.
[34] Z. Fang, K. Zhang, X. Zhang, B. Pan, Enhanced water decontamination from methylated arsenic by utilizing ultra-small hydrated zirconium oxides encapsulated inside gel-type anion exchanger, Chem. Eng. J. 430 (2022) 132641.
[35] G. Yan, T. Viraraghavan, M. Chen, A new model for heavy metal removal in a biosorption column, Adsorpt. Sci. Technol. 19 (2001) 25–43.
[36] A. Hethnawi, A.D. Manasrah, G. Vitale, N.N. Nassar, Fixed-bed column studies of total organic carbon removal from industrial wastewater by use of diatomite decorated with polyethylenimine-functionalized pyroxene nanoparticles, J. Colloid Interface Sci. 513 (2018) 28–42.
[37] T. Ataei-Germi, A. Nematollahzadeh, Bimodal porous silica microspheres decorated with polydopamine nano-particles for the adsorption of methylene blue in fixed-bed columns, J. Colloid Interface Sci. 470 (2016) 172–182.
[38] G. Alberti, V. Amendola, M. Pesavento, R. Biesuz, Beyond the synthesis of novel solid phases: review on modelling of sorption phenomena, Coord. Chem. Rev. 256 (2012) 28–45.
[39] A.A. Ghani, K.C. Devarayapalli, B. Kim, Y. Lim, G. Kim, J. Jang, D.S. Lee, Sodium-alginate-laden MXene and MOF systems and their composite hydrogel beads for batch and fixed-bed adsorption of naproxen with electrochemical regeneration, Carbohydr. Polym. 318 (2023) 121098.
[40] J. Staudt, F.B. Scheufele, C. Ribeiro, T.Y. Sato, R. Canevesi, C.E. Borba, Ciprofloxacin desorption from gel type ion exchange resin: desorption modeling in batch system and fixed bed column, Sep. Purif. Technol. 230 (2020) 115857.
[41] N.C. Fontão, F.V. Hackbarth, D.A. Mayer, L.P. Mazur, A.A.U. de Souza, V.J. P. Vilar, S.M.A.G.U. de Souza, A step forward on mathematical modeling of barium removal from aqueous solutions using seaweeds as natural cation exchangers: batch and fixed-bed systems, Chem. Eng. J. 401 (2020) 126019.
[42] A.G. Rios, A.M. Ribeiro, A.E. Rodrigues, A.F.P. Ferreira, Bovine serum albumin and myoglobin separation by size exclusion SMB, J. Chromatogr. A 1628 (2020) 461431.
[43] A. Hethnawi, N.N. Nassar, A.D. Manasrah, G. Vitale, Polyethylenimine-functionalized pyroxene nanoparticles embedded on Diatomite for adsorptive removal of dye from textile wastewater in a fixed-bed column, Chem. Eng. J. 320 (2017) 389–404.
[44] D.M. Ruthven, Principles of Adsorption and Adsorption Processes, John Wiley & Sons, New York, 1984.
[45] J.T. Edward, Molecular volumes and the Stokes-Einstein equation, J. Chem. Educ. 47 (1970) 261.
[46] Z. Xu, J. Cai, B. Pan, Mathematically modeling fixed-bed adsorption in aqueous systems, J. Zhejiang Univ.-Sci. A 14 (2013) 155–176.
[47] Q. Hu, D. Wang, S. Pang, L. Xu, Prediction of breakthrough curves for multicomponent adsorption in a fixed-bed column using logistic and Gompertz functions, Arab. J. Chem. 15 (2022) 104034.
[48] C. Rabbat, A. Pinna, Y. Andres, A. Villot, S. Awad, Adsorption of ibuprofen from aqueous solution onto a raw and steam-activated biochar derived from recycled textiles insulation panels at end-of-life: kinetic, isotherm and fixed-bed experiments, J. Water Process. Eng. 53 (2023) 103830.
[49] Y. Wang, C. Wang, X. Huang, Q. Zhang, T. Wang, X. Guo, Guideline for modeling solid-liquid adsorption: kinetics, isotherm, fixed bed, and thermodynamics, Chemosphere 349 (2024) 140736.
[50] E. Worch, Fixed-bed adsorption in drinking water treatment: a critical review on models and parameter estimation, J. Water Supply Res Technol.-Aqua 57 (2008) 171–183.
[51] B.O. Fagbayigbo, B.O. Opeolu, O.S. Fatoki, Adsorption of perfluorooctanoic acid (PFOA) and perfluorooctane sulfonate (PFOS) from water using leaf biomass (*Vitis vinifera*) in a fixed-bed column study, J. Environ. Health Sci. Eng. 18 (2020) 221–233.
[52] K. Al-Zawahreh, M.T. Barral, Y. Al-Degs, R. Paradelo, Competitive removal of textile dyes from solution by pine bark-compost in batch and fixed bed column experiments, Environ. Technol. Innov. 27 (2022) 102421.
[53] P. Ostaszewski, O. Długosz, M. Banach, Analysis of measuring methods of the concentration of methylene blue in the sorption process in fixed-bed column, Int. J. Environ. Sci. Technol. 19 (2022) 1–8.
[54] Q. Hu, Z. Zhang, Comment on Exponential and logistic functions: the two faces of the Bohart–Adams model, J. Hazard. Mater. 394 (2020) 122508.
[55] Q. Hu, Z. Zhang, Comment on Breakthrough curve analysis by simplistic models of fixed bed adsorption: in defense of the century-old Bohart–Adams model, Chem. Eng. J. 394 (2020) 124511.
[56] S. Wu, L. Debnath, A generalization of L'Hôspital-type rules for monotonicity and its application, Appl. Math. Lett. 22 (2009) 284–290.
[57] R.A. Hutchins, New method simplifies design of activated carbon systems, Chem. Eng. 80 (1973) 133–138.
[58] R. Han, Y. Wang, W. Zou, Y. Wang, J. Shi, Comparison of linear and nonlinear analysis in estimating the Thomas model parameters for methylene blue

adsorption onto natural zeolite in fixed-bed column, J. Hazard. Mater. 145 (2007) 331–335.
[59] K. Xiao, X. Wang, X. Huang, T.D. Waite, X. Wen, Analysis of polysaccharide, protein and humic acid retention by microfiltration membranes using Thomas' dynamic adsorption model, J. Membr. Sci. 342 (2009) 22–34.
[60] R. Lakshmipathy, N.C. Sarada, A fixed bed column study for the removal of Pb^{2+} ions by watermelon rind, Environ. Sci. Water Res. Technol. 1 (2015) 244–250.
[61] D. Ranjan Rout, H. Mohan Jena, Synthesis of novel reduced graphene oxide decorated β-cyclodextrin epichlorohydrin composite and its application for Cr(VI) removal: batch and fixed-bed studies, Sep. Purif. Technol. 278 (2021) 119630.
[62] K.H. Chu, Fixed bed adsorption of water contaminants: a cautionary guide to simple analytical models and modeling misconceptions, Sep. Purif. Rev. 52 (2023) 75–97.
[63] H. Xiang, H. Zhang, P. Liu, Y. Yan, Adsorption dynamics of ethane from air in structured fixed beds with different microfibrous composites, Chin. J. Chem. Eng. 53 (2023) 14–24.
[64] Q. Hu, Y. Xie, Z. Zhang, Modification of breakthrough models in a continuous-flow fixed-bed column: mathematical characteristics of breakthrough curves and rate profiles, Sep. Purif. Technol. 238 (2020) 116399.
[65] Y.H. Yoon, J.H. Nelson, Application of gas adsorption kinetics — II. A theoretical model for respirator cartridge service life and its practical applications, Am. Ind. Hyg. Assoc. J. 45 (1984) 517–524.
[66] J.T. de Oliveira, L.R. de Carvalho Costa, G.D. Agnol, L.A. Féris, Experimental design and data prediction by Bayesian statistics for adsorption of tetracycline in a GAC fixed-bed column, Sep. Purif. Technol. 319 (2023) 124097.
[67] Q. Hu, R. Lan, L. He, H. Liu, X. Pei, A critical review of adsorption isotherm models for aqueous contaminants: curve characteristics, site energy distribution and common controversies, J. Environ. Manag. 329 (2023) 117104.
[68] U. Kumari, A. Mishra, H. Siddiqi, B.C. Meikap, Effective defluoridation of industrial wastewater by using acid modified alumina in fixed-bed adsorption column: experimental and breakthrough curves analysis, J. Clean. Prod. 279 (2021) 123645.
[69] A.A. Aryee, R. Han, A novel biocomposite based on peanut husk with antibacterial properties for the efficient sequestration of trimethoprim in solution: batch and column adsorption studies, Colloids Surf. A Physicochem. Eng. Asp. 635 (2022) 128051.
[70] M. Vera, D.M. Juela, C. Cruzat, E. Vanegas, Modeling and computational fluid dynamic simulation of acetaminophen adsorption using sugarcane bagasse, J. Environ. Chem. Eng. 9 (2021) 105056.
[71] A. Wolborska, Adsorption on activated carbon of p-nitrophenol from aqueous solution, Water Res. 23 (1989) 85–91.
[72] C. Smaranda, M.C. Popescu, D. Bulgariu, T. Măluţan, M. Gavrilescu, Adsorption of organic pollutants onto a Romanian soil: column dynamics and transport, Process. Saf. Environ. Prot. 108 (2017) 108–120.
[73] S. Singh, V.C. Srivastava, I.D. Mall, Fixed-bed study for adsorptive removal of furfural by activated carbon, Colloids Surf. A Physicochem. Eng. Asp. 332 (2009) 50–56.
[74] A. Wolborska, P. Pustelnik, A simplified method for determination of the break-through time of an adsorbent layer, Water Res. 30 (1996) 2643–2650.
[75] A. Katsigiannis, C. Noutsopoulos, J. Mantziaras, M. Gioldasi, Removal of emerging pollutants through granular activated carbon, Chem. Eng. J. 280 (2015) 49–57.
[76] D. Sana, S. Jalila, A comparative study of adsorption and regeneration with different agricultural wastes as adsorbents for the removal of methylene blue from aqueous solution, Chin. J. Chem. Eng. 25 (2017) 1282–1287.
[77] C.G. Lee, J.H. Kim, J.K. Kang, S.B. Kim, S.J. Park, S.H. Lee, J.W. Choi, Comparative analysis of fixed-bed sorption models using phosphate breakthrough curves in slag filter media, Desalin. Water Treat. 55 (2015) 1795–1805.
[78] A. Klinkenberg, Numerical evaluation of equations describing transient heat and mass transfer in packed solids, Ind. Eng. Chem. 40 (1948) 1992–1994.
[79] Y. Taamneh, R. Al Dwairi, The efficiency of Jordanian natural zeolite for heavy metals removal, Appl Water Sci 3 (2013) 77–84.
[80] K.H. Chu, Prediction of arsenic breakthrough in a pilot column of polymer-supported nanoparticles, J. Water Process. Eng. 3 (2014) 117–122.
[81] B. Ren, N. Lyczko, Y. Zhao, A. Nzihou, Alum sludge as an efficient sorbent for hydrogen sulfide removal: experimental, mechanisms and modeling studies, Chemosphere 248 (2020) 126010.
[82] J.M. Silva, M.F. Ribeiro, I. Graça, A. Fernandes, Bio-oils/FCC co-processing: Insights into the adsorption of guaiacol on Y zeolites with distinct acidity and textural properties, Microporous Mesoporous Mater. 323 (2021) 111170.
[83] J.B. Dima, M. Ferrari, N. Zaritzky, Mathematical modeling of fixed-bed columns adsorption: hexavalent chromium onto chitosan flakes, Ind. Eng. Chem. Res. 59 (2020) 15378–15386.
[84] J.M. Chern, Y.W. Chien, Adsorption of nitrophenol onto activated carbon: isotherms and breakthrough curves, Water Res. 36 (2002) 647–655.
[85] J.M. Chern, Y.W. Chien, Competitive adsorption of benzoic acid and p-nitrophenol onto activated carbon: isotherm and breakthrough curves, Water Res. 37 (2003) 2347–2356.
[86] B.C. Pan, F.W. Meng, X.Q. Chen, B.J. Pan, X.T. Li, W.M. Zhang, X. Zhang, J. L. Chen, Q.X. Zhang, Y. Sun, Application of an effective method in predicting breakthrough curves of fixed-bed adsorption onto resin adsorbent, J. Hazard. Mater. 124 (2005) 74–80.
[87] B. Pan, X. Chen, B. Pan, W. Zhang, X. Zhang, Q. Zhang, Preparation of an aminated macroreticular resin adsorbent and its adsorption of p-nitrophenol from water, J. Hazard. Mater. 137 (2006) 1236–1240.
[88] X. Zhang, S. Chen, H.T. Bi, Application of wave propagation theory to adsorption breakthrough studies of toluene on activated carbon fiber beds, Carbon 48 (2010) 2317–2326.
[89] A. Ararem, A. Bouzidi, B. Mohamedi, O. Bouras, Modeling of fixed-bed adsorption of Cs^{+} and Sr^{2+} onto clay–iron oxide composite using artificial neural network and constant–pattern wave approach, J. Radioanal. Nucl. Chem. 301 (2014) 881–887.
[90] E.F. Schuster, Estimating the distribution function of a symmetric distribution, Biometrika 62 (1975) 631–635.
[91] C.E. Borba, R. Guirardello, E.A. Silva, M.T. Veit, C.R.G. Tavares, Removal of nickel(II) ions from aqueous solution by biosorption in a fixed bed column: experimental and theoretical breakthrough curves, Biochem. Eng. J. 30 (2006) 184–191.
[92] A. Karami, R. Sabouni, M.H. Al-Sayah, A. Aidan, Adsorption potentials of iron-based metal–organic framework for methyl orange removal: batch and fixed-bed column studies, Int. J. Environ. Sci. Technol. 18 (2021) 3597–3612.
[93] M.I. El-Khaiary, Least-squares regression of adsorption equilibrium data: comparing the options, J. Hazard. Mater. 158 (2008) 73–87.
[94] C.M.B. de Araujo, G. Wernke, M.G. Ghislandi, A. Diório, M.F. Vieira, R. Bergamasco, M.A. da Motta Sobrinho, A.E. Rodrigues, Continuous removal of pharmaceutical drug chloroquine and Safranin-O dye from water using agar-graphene oxide hydrogel: selective adsorption in batch and fixed-bed experiments, Environ. Res. 216 (2023) 114425.
[95] A.K.A. Khalil, I.W. Almanassra, A. Chatla, I. Ihsanullah, T. Laoui, M. Ali Atieh, Insights into the adsorption of lead ions by Mg-Al LDH doped activated carbon composites: implications for fixed bed column and batch applications, Chem. Eng. Sci. 281 (2023) 119192.
[96] R. Kopelman, Fractal reaction kinetics, Science 241 (1988) 1620–1626.
[97] Q. Hu, H. Liu, Z. Zhang, X. Pei, Development of fractal-like Clark model in a fixed-bed column, Sep. Purif. Technol. 251 (2020) 117396.
[98] R. Kopelman, Rate processes on fractals: theory, simulations, and experiments, J. Stat. Phys. 42 (1986) 185–200.
[99] M. Balsamo, F. Montagnaro, Liquid–solid adsorption processes interpreted by fractal-like kinetic models, Environ. Chem. Lett. 17 (2019) 1067–1075.
[100] M. Haerifar, S. Azizian, Fractal-like adsorption kinetics at the solid/solution interface, J. Phys. Chem. C 116 (2012) 13111–13119.
[101] M. Haerifar, S. Azizian, Fractal-like kinetics for adsorption on heterogeneous solid surfaces, J. Phys. Chem. C 118 (2014) 1129–1134.
[102] H. Bashiri, A. Shajari, Theoretical study of fractal-like kinetics of adsorption, Adsorpt. Sci. Technol. 32 (2014) 623–634.
[103] F. Montagnaro, M. Balsamo, Modelling CO_2 adsorption dynamics onto amine-functionalised sorbents: a fractal-like kinetic perspective, Chem. Eng. Sci. 192 (2018) 603–612.
[104] Q. Hu, S. Pang, D. Wang, Y. Yang, H. Liu, Deeper insights into the Bohart–Adams model in a fixed-bed column, J. Phys. Chem. B 125 (2021) 8494–8501.
[105] C.K. Rojas-Mayorga, A. Bonilla-Petriciolet, F.J. Sánchez-Ruiz, J. Moreno-Pérez, H. E. Reynel-Ávila, I.A. Aguayo-Villarreal, D.I. Mendoza-Castillo, Breakthrough curve modeling of liquid-phase adsorption of fluoride ions on aluminum-doped bone char using micro-columns: effectiveness of data fitting approaches, J. Mol. Liq. 208 (2015) 114–121.
[106] N. Blagojev, D. Kukić, V. Vasić, M. Šćiban, J. Prodanović, O. Bera, A new approach for modelling and optimization of Cu(II) biosorption from aqueous solutions using sugar beet shreds in a fixed-bed column, J. Hazard. Mater. 363 (2019) 366–375.
[107] N. Blagojev, V. Vasić, D. Kukić, M. Šćiban, J. Prodanović, O. Bera, Modelling and efficiency evaluation of the continuous biosorption of Cu(II) and Cr(VI) from water by agricultural waste materials, J. Environ. Manag. 281 (2021) 111876.
[108] Q. Hu, Y. Xie, C. Feng, Z. Zhang, Prediction of breakthrough behaviors using logistic, hyperbolic tangent and double exponential models in the fixed-bed column, Sep. Purif. Technol. 212 (2019) 572–579.
[109] K.H. Chu, Fitting the Gompertz equation to asymmetric breakthrough curves, J. Environ. Chem. Eng. 8 (2020) 103713.
[110] W. Weibull, A statistical distribution function of wide applicability, J. Appl. Mech. 293–297 (1951).
[111] K.H. Chu, Fixed bed adsorption of chromium and the Weibull function, J. Hazard. Mater. Lett. 2 (2021) 100022.
[112] R. Apiratikul, K.H. Chu, Improved fixed bed models for correlating asymmetric adsorption breakthrough curves, J. Water Process. Eng. 40 (2021) 101810.
[113] K.H. Chu, M.A. Hashim, Fixed bed adsorption of water and air contaminants: analysis of breakthrough curves using probability distribution functions, Chem. Eng. Commun. 210 (2023) 1528–1537.
[114] J. Singh, S.K. Kumaresan, S. Swaroop, V. Mishra, Development of predictive model for the fixed-bed column reactor, Appl Water Sci 13 (2023) 114.
[115] P.K. Kimani, Asymmetrical fixed-bed breakthrough curve modelling: comparing simplistic, log-modified, fractal-like, and probability distribution function models, Chem. Eng. Res. Des. 201 (2024) 446–456.
[116] M.F. Oliveira, V.M. de Souza, M.G.C. da Silva, M.G.A. Vieira, Fixed-bed adsorption of caffeine onto thermally modified Verde-lodo bentonite, Ind. Eng. Chem. Res. 57 (2018) 17480–17487.
[117] M.I. El-Khaiary, G.F. Malash, Common data analysis errors in batch adsorption studies, Hydrometallurgy 105 (2011) 314–320.
[118] A. Scott, C. Wild, Transformations and R^2, Am. Stat. 45 (1991) 127–129.
[119] A. Bakka, R. Mamouni, N. Saffaj, A. Laknifli, K. Aziz, A. Roudani, Removal of bifenthrin pesticide from aqueous solutions by treated patellidae shells using a new fixed bed column filtration technique, Process. Saf. Environ. Prot. 143 (2020) 55–65.

[120] C.C. Chen, R. Vance Morey, Comparison of four EMC/ERH equations, Trans. ASAE 32 (1989) 983–0990.

[121] M.B. de Farias, M.P. Spaolonzi, M.G.C. Silva, M.G.A. Vieira, Fixed-bed adsorption of bisphenol A onto organoclay: characterisation, mathematical modelling and theoretical calculation of DFT-based chemical descriptors, J. Environ. Chem. Eng. 9 (2021) 106103.

[122] L.S. Roca, S.E. Schoemaker, B.W.J. Pirok, A.F.G. Gargano, P.J. Schoenmakers, Accurate modelling of the retention behaviour of peptides in gradient-elution hydrophilic interaction liquid chromatography, J. Chromatogr. A 1614 (2020) 460650.

[123] G.F. Malash, M.I. El-Khaiary, Piecewise linear regression: a statistical method for the analysis of experimental adsorption data by the intraparticle-diffusion models, Chem. Eng. J. 163 (2010) 256–263.

[124] G. Glatting, P. Kletting, S.N. Reske, K. Hohl, C. Ring, Choosing the optimal fit function: comparison of the Akaike information criterion and the *F*-test, Med. Phys. 34 (2007) 4285–4292.

[125] S. Cataldo, A. Gianguzza, D. Milea, N. Muratore, A. Pettignano, S. Sammartano, A critical approach to the toxic metal ion removal by hazelnut and almond shells, Environ. Sci. Pollut. Res. 25 (2018) 4238–4253.

[126] T.M. Darweesh, M.J. Ahmed, Adsorption of ciprofloxacin and norfloxacin from aqueous solution onto granular activated carbon in fixed bed column, Ecotoxicol. Environ. Saf. 138 (2017) 139–145.

[127] H. Bacelo, S.C.R. Santos, A. Ribeiro, R.A.R. Boaventura, C.M.S. Botelho, Antimony removal from water by pine bark tannin resin: Batch and fixed-bed adsorption, J. Environ. Manag. 302 (2022) 114100.

[128] D. Juela, M. Vera, C. Cruzat, A. Astudillo, E. Vanegas, A new approach for scaling up fixed-bed adsorption columns for aqueous systems: a case of antibiotic removal on natural adsorbent, Process. Saf. Environ. Prot. 159 (2022) 953–963.

[129] Y. Lee, Y. Ren, M. Cui, Y. Zhou, O. Kwon, J. Ko, J. Khim, Arsenic adsorption study in acid mine drainage using fixed bed column by novel beaded adsorbent, Chemosphere 291 (2022) 132894.

[130] J. Jang, D.S. Lee, Effective phosphorus removal using chitosan/Ca-organically modified montmorillonite beads in batch and fixed-bed column studies, J. Hazard. Mater. 375 (2019) 9–18.

[131] M.E. González-López, C.M. Laureano-Anzaldo, A.A. Pérez-Fonseca, M. Arellano, J. R. Robledo-Ortíz, A critical overview of adsorption models linearization: methodological and statistical inconsistencies, Sep. Purif. Rev. 51 (2022) 358–372.

[132] A.L. Silveira Neto, W. Pimentel-Almeida, G. Niero, E.H. Wanderlind, C. M. Radetski, G.I. Almerindo, Application of a biochar produced from malt bagasse as a residue of brewery industry in fixed-bed column adsorption of paracetamol, Chem. Eng. Res. Des. 194 (2023) 779–786.

[133] J. Tejedor, R. Álvarez-Briceño, V.H. Guerrero, C.A. Villamar-Ayala, Removal of caffeine using agro-industrial residues in fixed-bed columns: improving the adsorption capacity and efficiency by selecting adequate physical and operational parameters, J. Water Process. Eng. 53 (2023) 103778.

[134] J. He, Q. Zhou, J. Guo, J. Gao, F. Fang, Incredulity on assumptions for the simplified Bohart-Adams model: 17a-ethinylestradiol separation in lab-scale anthracite columns, J. Hazard. Mater. 384 (2020) 121501.

[135] Y.S. Ho, J.F. Porter, G. McKay, Equilibrium isotherm studies for the sorption of divalent metal ions onto peat: copper, nickel and lead single component systems, Water Air Soil Pollut. 141 (2002) 1–33.

[136] K.H. Chen, Y.R. Lai, N.T.D. Hanh, Breakthrough curve modeling for lysozyme by ion-exchange nanofiber membrane: linear and nonlinear analysis, J. Taiwan Inst. Chem. Eng. 105198 (2023).

[137] H.N. Tran, S.J. You, A. Hosseini-Bandegharaei, H.P. Chao, Mistakes and inconsistencies regarding adsorption of contaminants from aqueous solutions: a critical review, Water Res. 120 (2017) 88–116a.

[138] Z. Ma, R.D. Whitley, N.H.L. Wang, Pore and surface diffusion in multicomponent adsorption and liquid chromatography systems, AICHE J. 42 (1996) 1244–1262.